DEN

EXECUTIVE REFERENCE DATA

Energy Management Towards The National Energy Security

PREFACE

Indonesia's economic growth with an average growth rate of 5.7% per year during the period 2011 through 2012, has trigerred the increase of domestic energy demand by an average of 4.4% per year. By 2030 the energy demand is estimated to be three times the demand level in 2010, while the increase in energy production in 2030 is estimated to rise at twice that of the year 2010. This condition requires a good energy planning and management to ensure the supply of energy in the long term.

To provide an overview of the energy situation in Indonesia, The National Energy Council published the 2nd edition of the "Executive Reference Data: Energy Management towards National Energy Security". This 2nd edition booklet in 2013, illustrates the energy management of Indonesia up to 2012. It presents information on the oil, gas, coal and electricity condition, the progress of 10,000 MW accelerated development of the electric power generation plants phase I and phase II, the development of new and renewable energy, CO2 emissions from the energy users sector and of the Indonesia’s energy position in the ASEAN region.

Hopefully this booklet provide a great benefits for energy management in Indonesia.

Jakarta, December 2013

Secretary General of
National Energy Council

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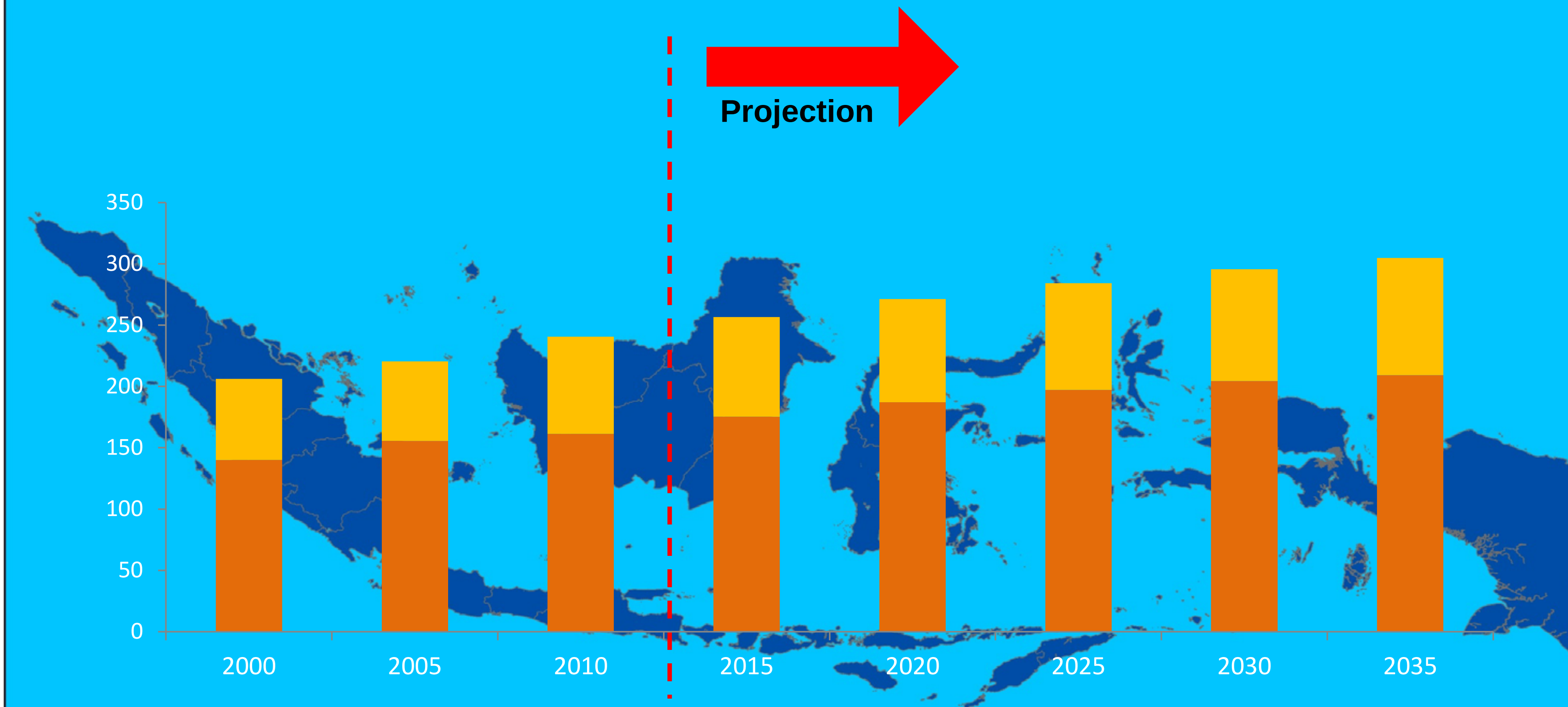
SOCIO-ECONOMIC

POPULATION

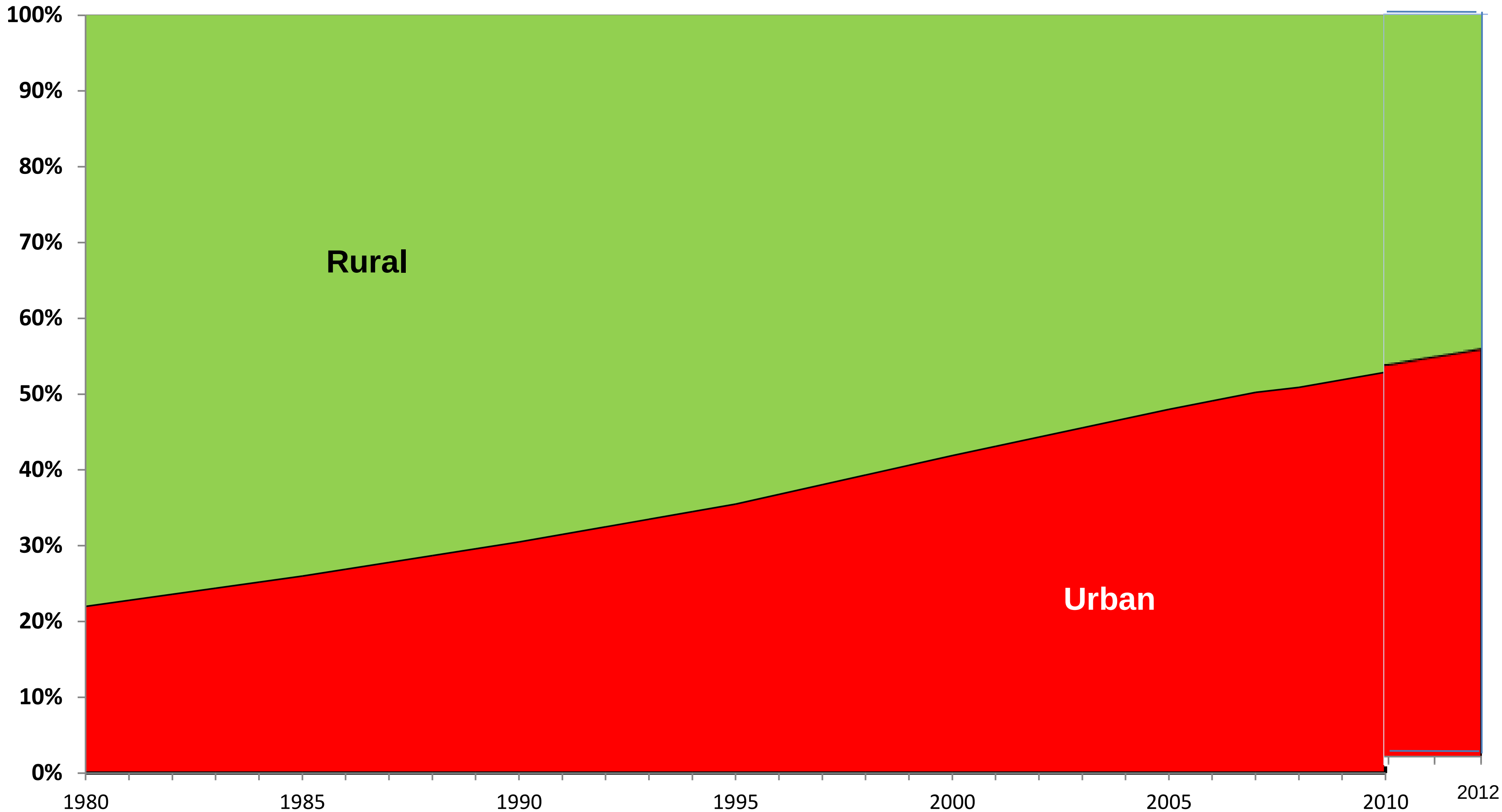
(Million peoples)



Since last 25 years, the numbers of population increase from 240,7 million in 2010 to 304,9 million in 2035, although average growth of Indonesia population in 2010-2035 shown the decreasing trends. In 2010-2015 and 2030-2035 , population growth rate decreased from 1,29 percent to 0,62 percent each year.

(Source: Indonesian Central Bureau of Statistics)

POPULATION IN RURAL AND URBAN AREAS

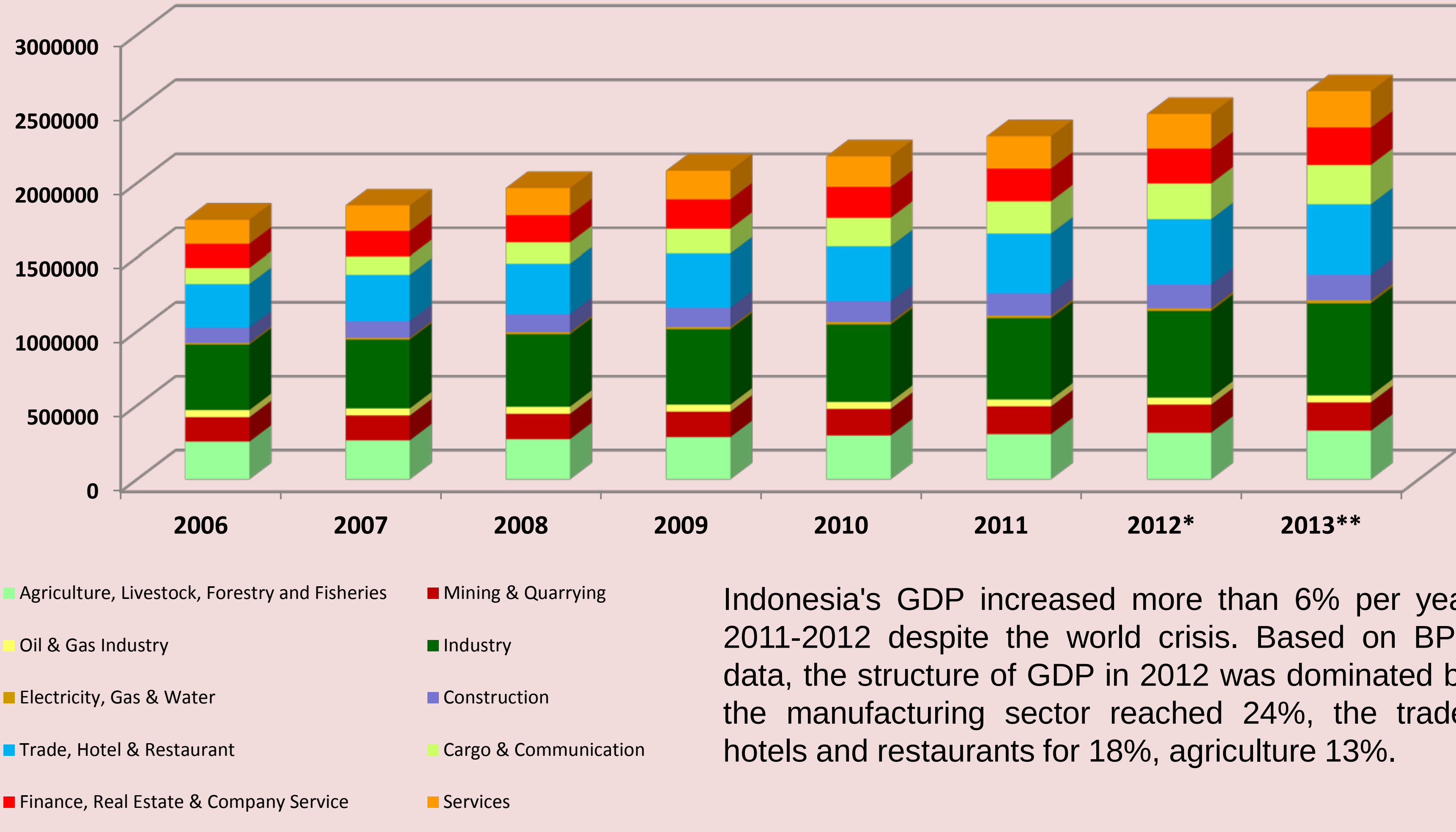


(Source: Bureau of Statistics Indonesia)

In 2015, it is estimated the population in the urban areas will increase with the composition of 56% living in town/city and 44% living in villages. Changes of a village to become a town has caused the increase in number and density of population, economic activity is no longer relies on the agricultural sector, to the improvement of infrastructure.

GDP CONSTANT 2000 BY SECTOR

(Billion IDR)



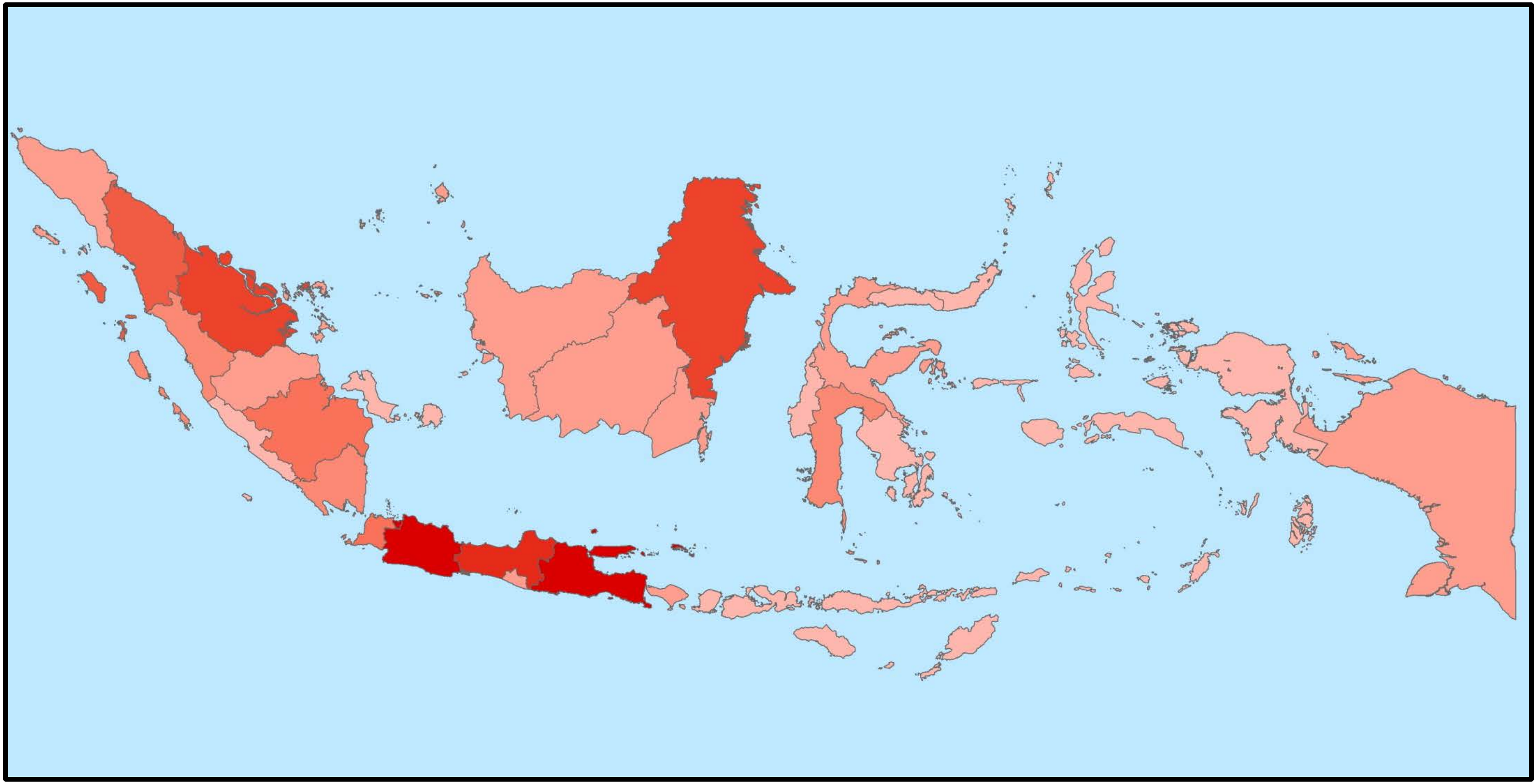
* Temporary Value ** Very Temporary Value

Indonesia's GDP increased more than 6% per year 2011-2012 despite the world crisis. Based on BPS data, the structure of GDP in 2012 was dominated by the manufacturing sector reached 24%, the trade, hotels and restaurants for 18%, agriculture 13%.

Source: BPS - *Statistics Indonesia*

GDP BY REGION

(GDPR in Billion IDR)

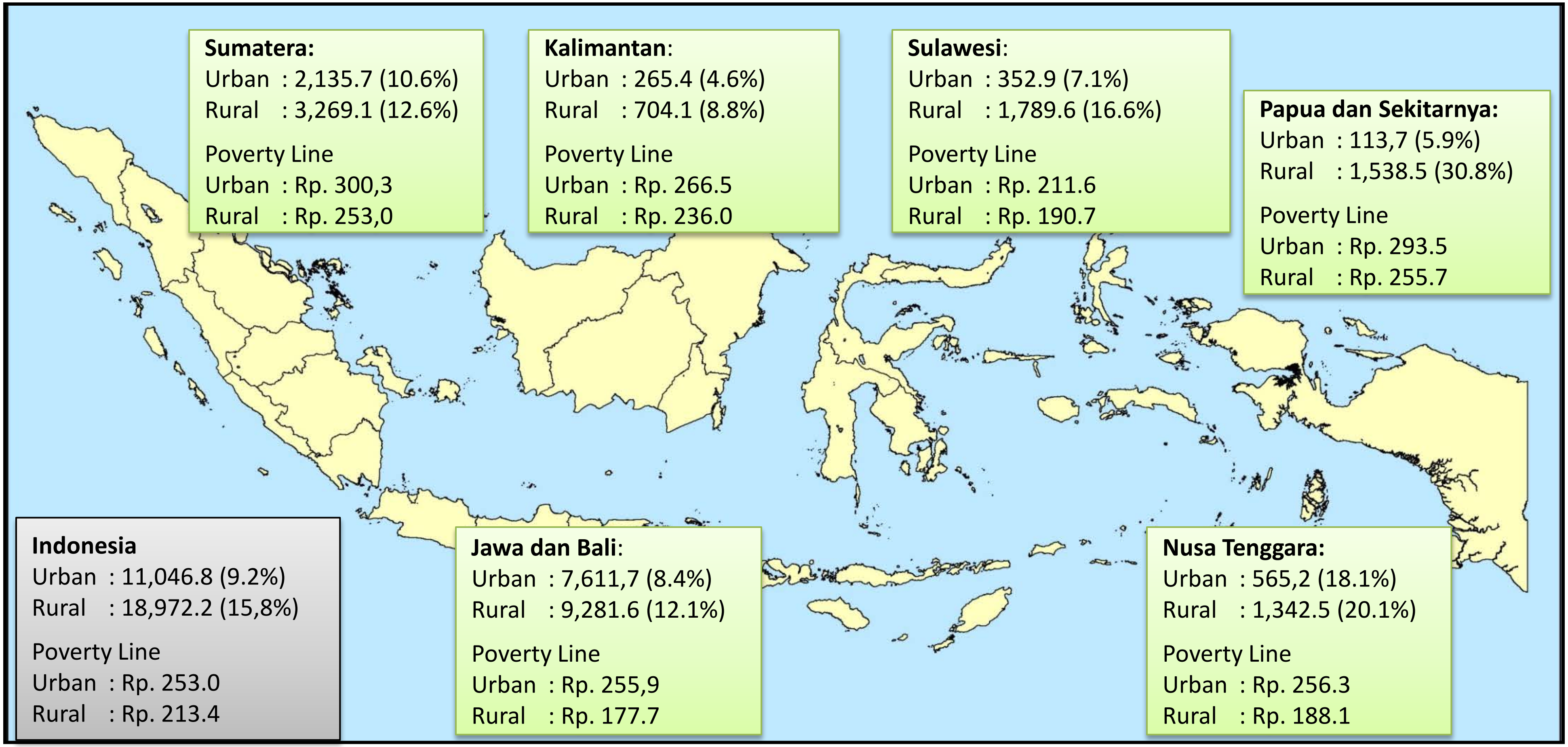


Source: BPS - *Statistics Indonesia*

- > 600.000
- 500.001 – 600.000
- 400.001 – 500.000
- 300.001 – 400.000
- 200.001 – 300.000
- 100.001 – 200.000
- 50.000 – 100.000
- < 50.000

POOR PEOPLE AND POVERTY BORDER LINE

(Thousand Peoples)



Source: BPS - *Statistics Indonesia, reprocessed by National Energy Council*

By 2012, around 11.04 million peoples living in urban areas where 9.2% are the poor. While the number of people left in rural areas was 18.97 million people, of which 15.8% of the total were categorized as the poor. The urban poverty line level was 253 thousand rupiahs, and in the rural area was 213 thousand rupiahs.

NUMBER OF MOTOR VEHICLES BY TYPES

Passenger Car

Subsidized fuel consumption for passenger cars are very huge, total fuel which subsidized by the Government almost 53 % consumed by passenger cars, and 40 % by motor cycles.

Bus

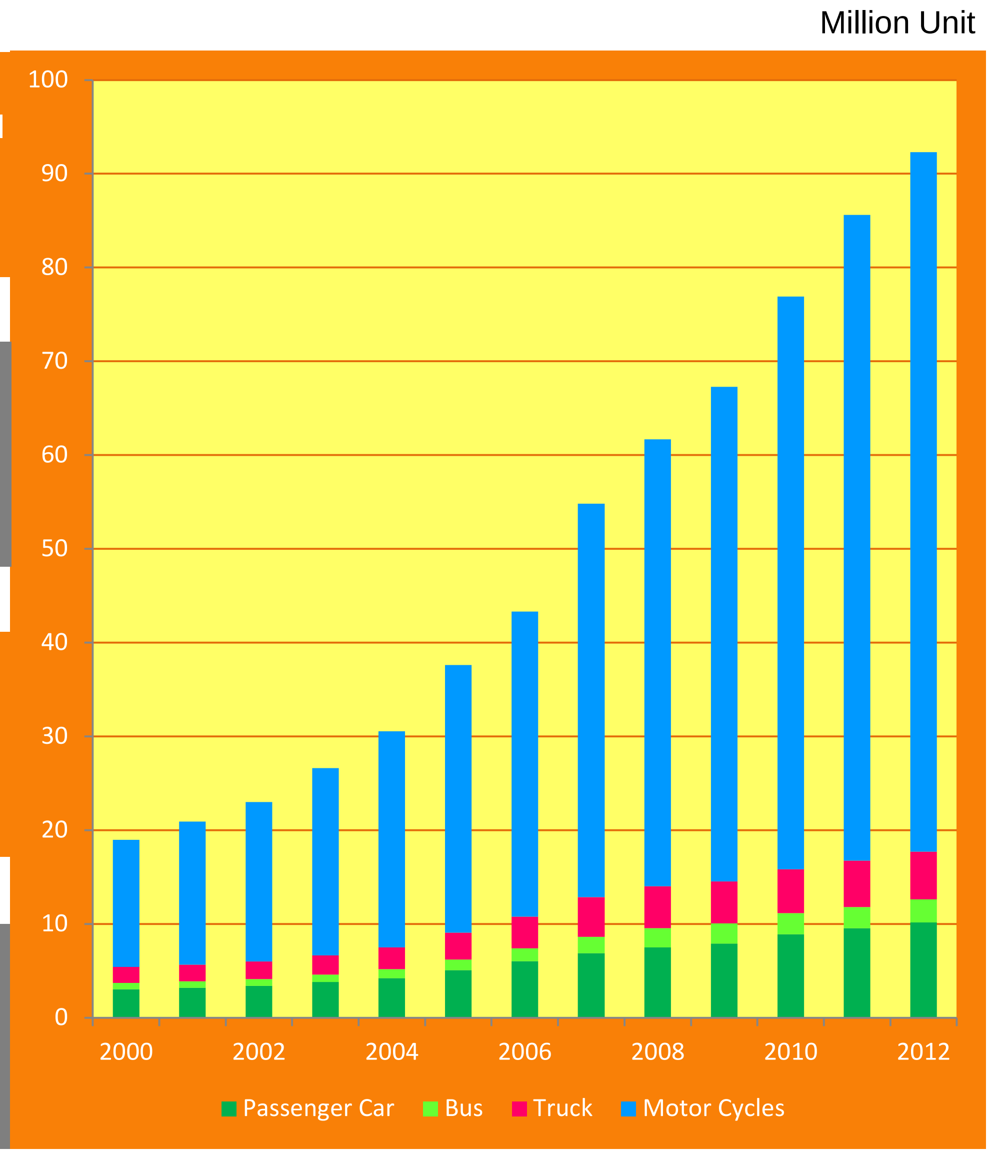
Bus consumed 3 % from total fuel which subsidized by government.

Truck

Truck consumed 4% from total fuel which subsidized by government

Motor Cycles

Motor cycles growth rate is very high (15.3% per year), exceed the growth rate of total vehicles by 14% per year.

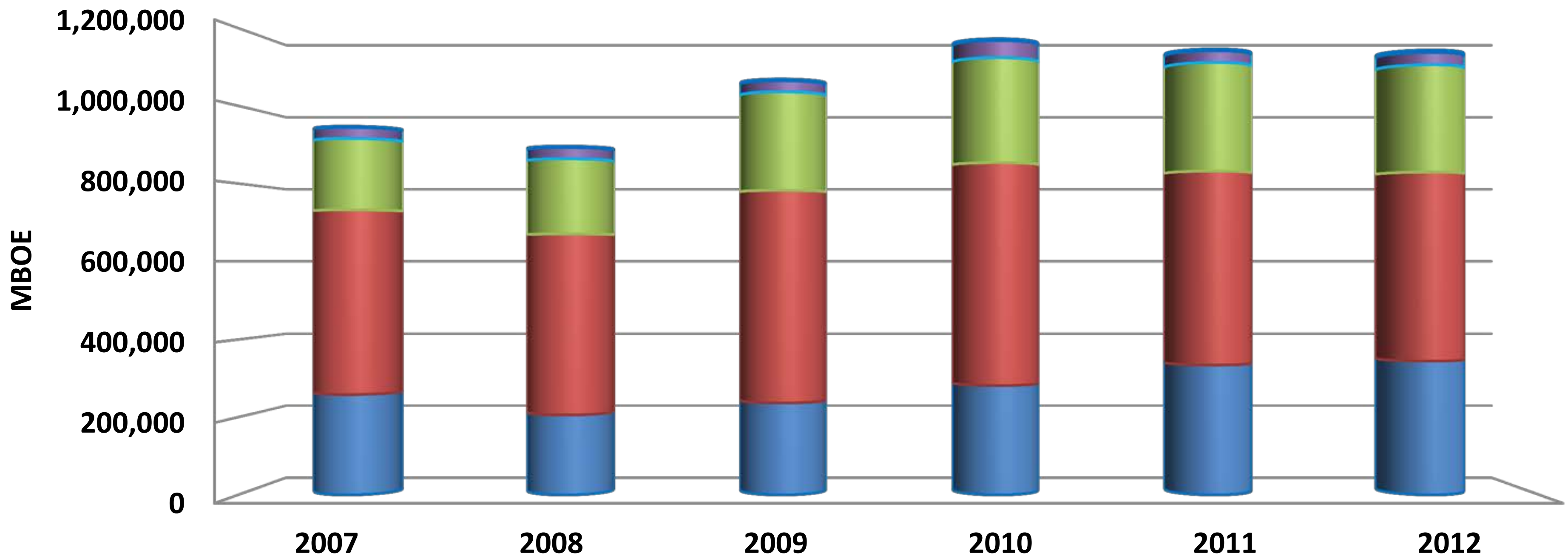


Source: Ministry of Transportation

ENERGY INDICATOR

PRIMARY ENERGY SUPPLY

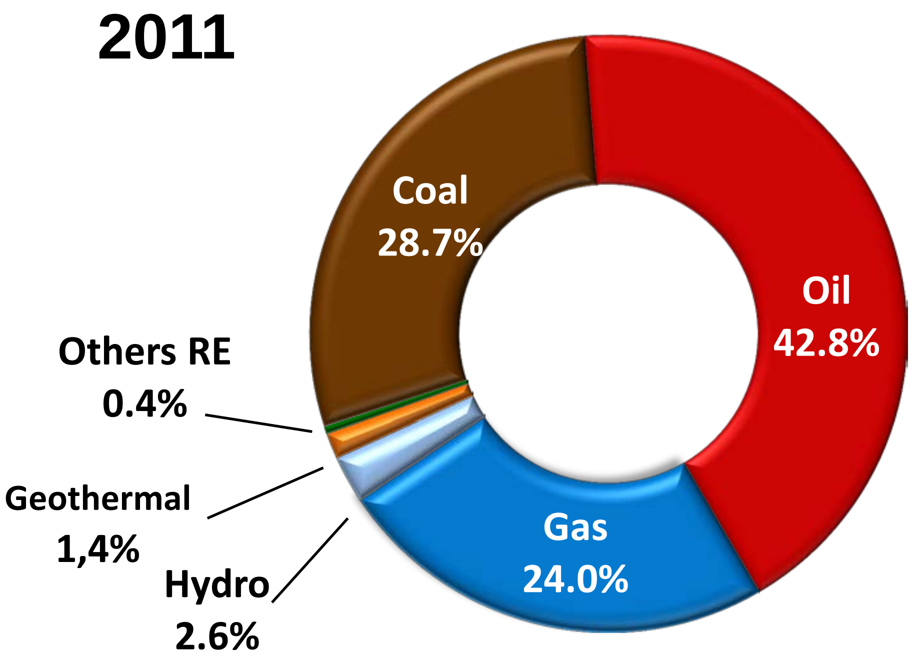
(MBOE)

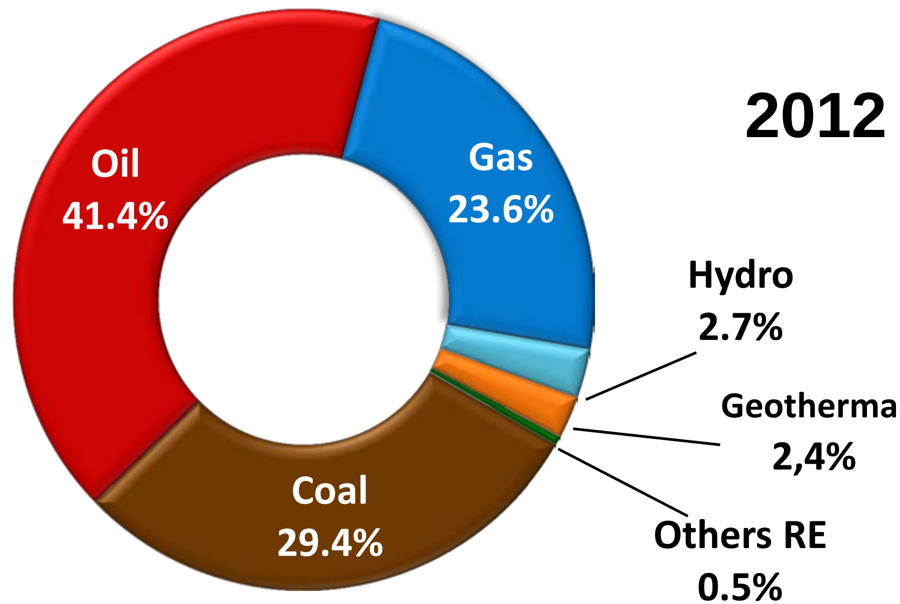


	2007	2008	2009	2010	2011	2012
Others RE	2,873	3,193	3,991	4,434	4,668	6,025
Hydro	28,451	29,060	28,696	44,003	30,074	31,795
Gas	183,624	193,352	253,198	269,942	279,286	275,950
Oil	474,033	464,704	546,050	572,249	497,729	485,045
Coal	258,174	205,492	236,439	281,400	334,143	344,400

Source: Ministry of Energy and Mineral Resources/MEMR, *reprocessed by NEC)*
Excluded Biomass

ENERGY MIX

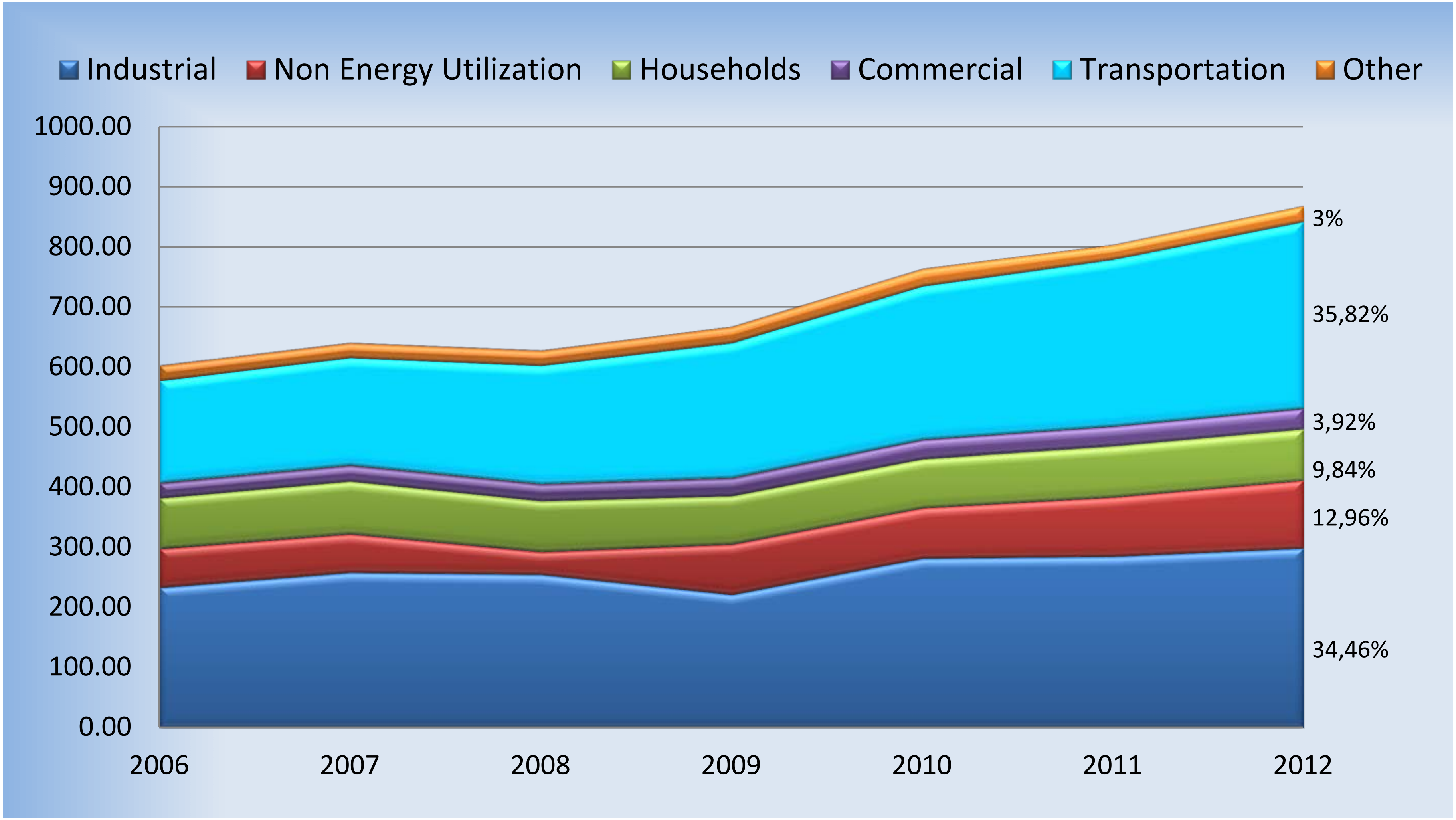




Source: MEMR, reprocessed by NEC

FINAL ENERGY CONSUMPTION BY SECTOR

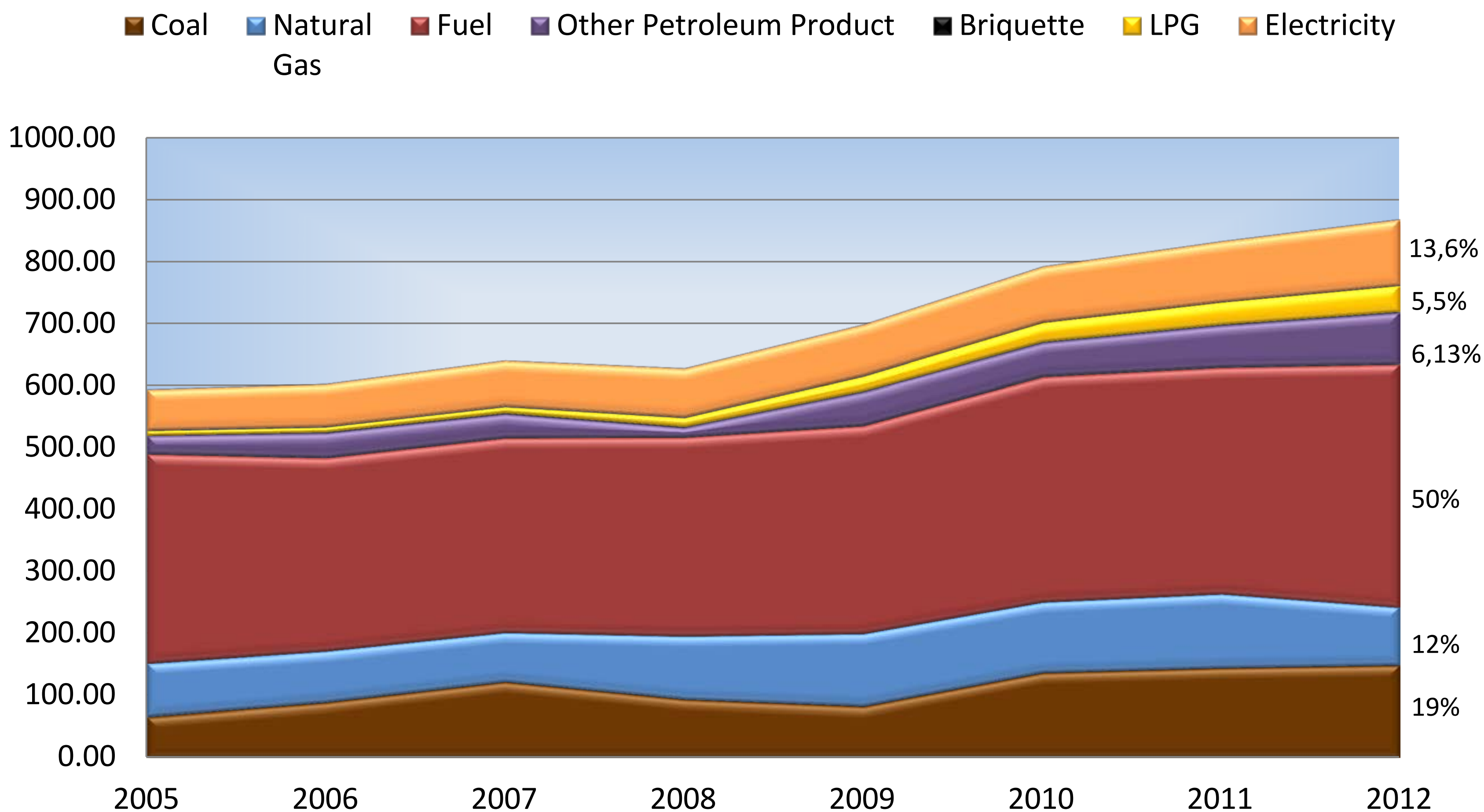
(MBOE)



Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*
Excluded Biomass

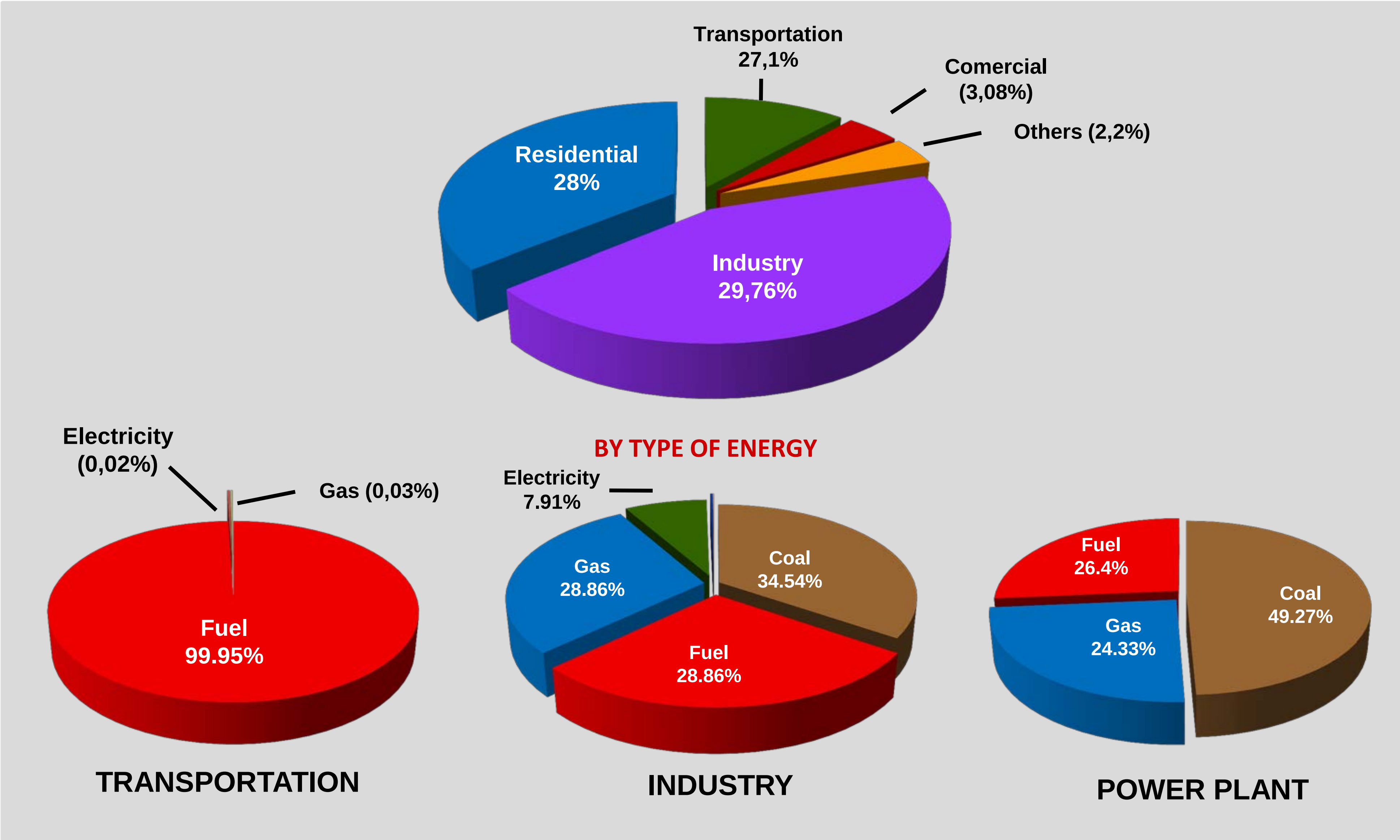
FINAL ENERGY CONSUMPTION

(MBOE)



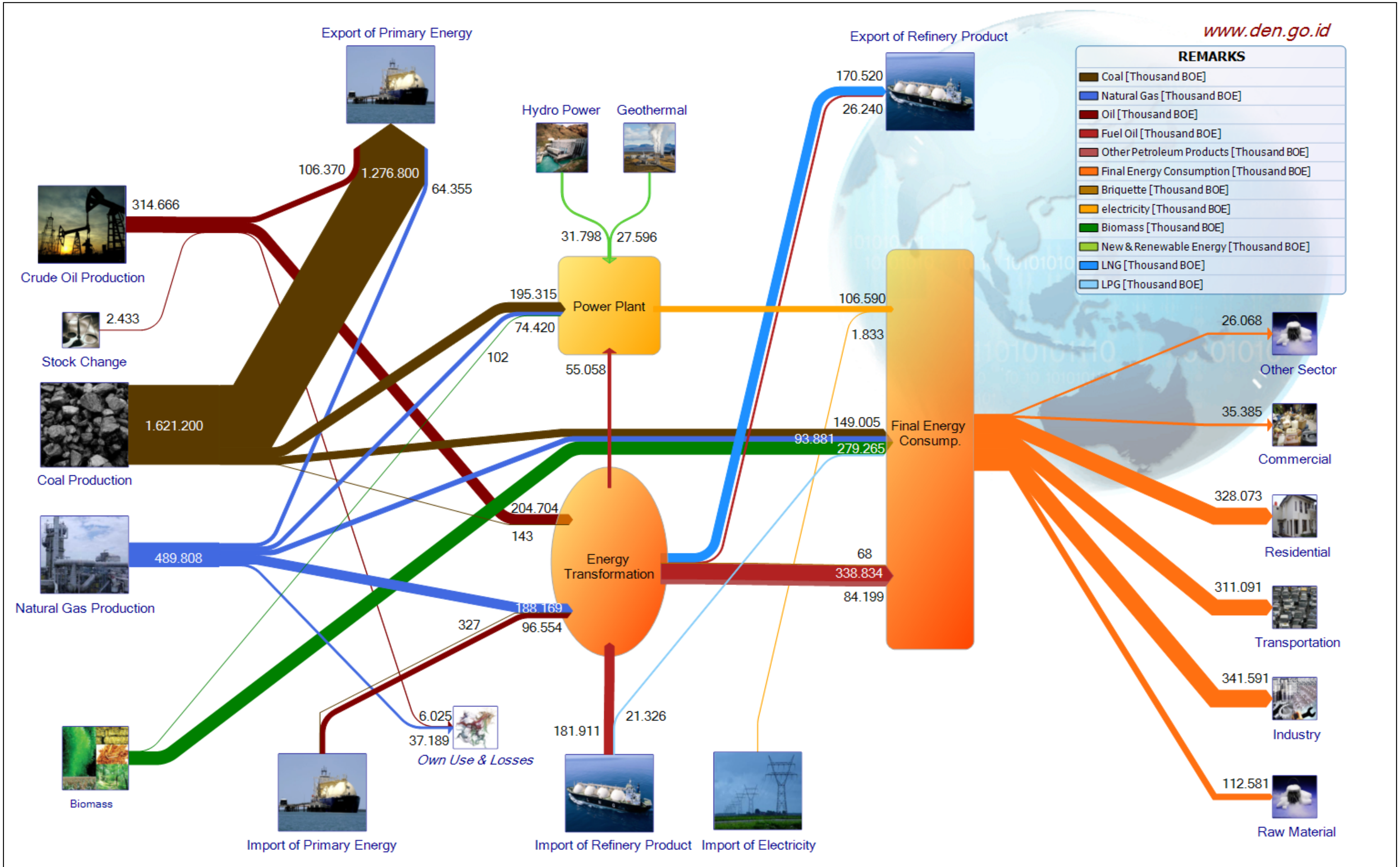
Source: Ministry of Energy and Mineral Resources, *reprocessed by NEC*

SHARES OF ENERGY UTILIZATION BY SECTOR



Source: Ministry of Energy and Mineral Resources , *"Handbook of Energy & Economic Statistic Indonesia 2012"*.

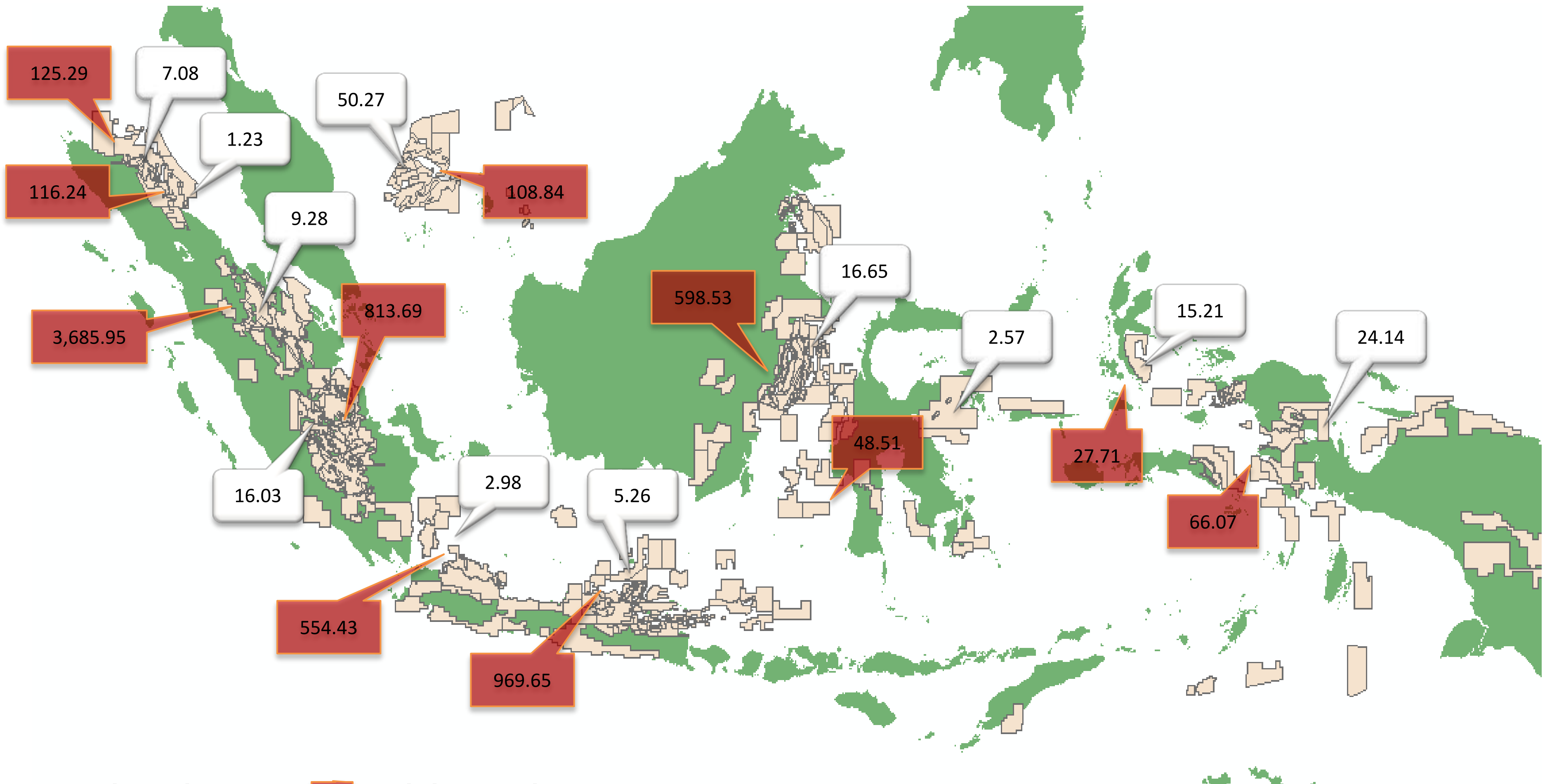
ENERGY FLOW 2012



Source: National Energy Council

OIL AND GAS

OIL AND GAS RESERVES



: Gas (TSCF) : Oil (MMSTB)

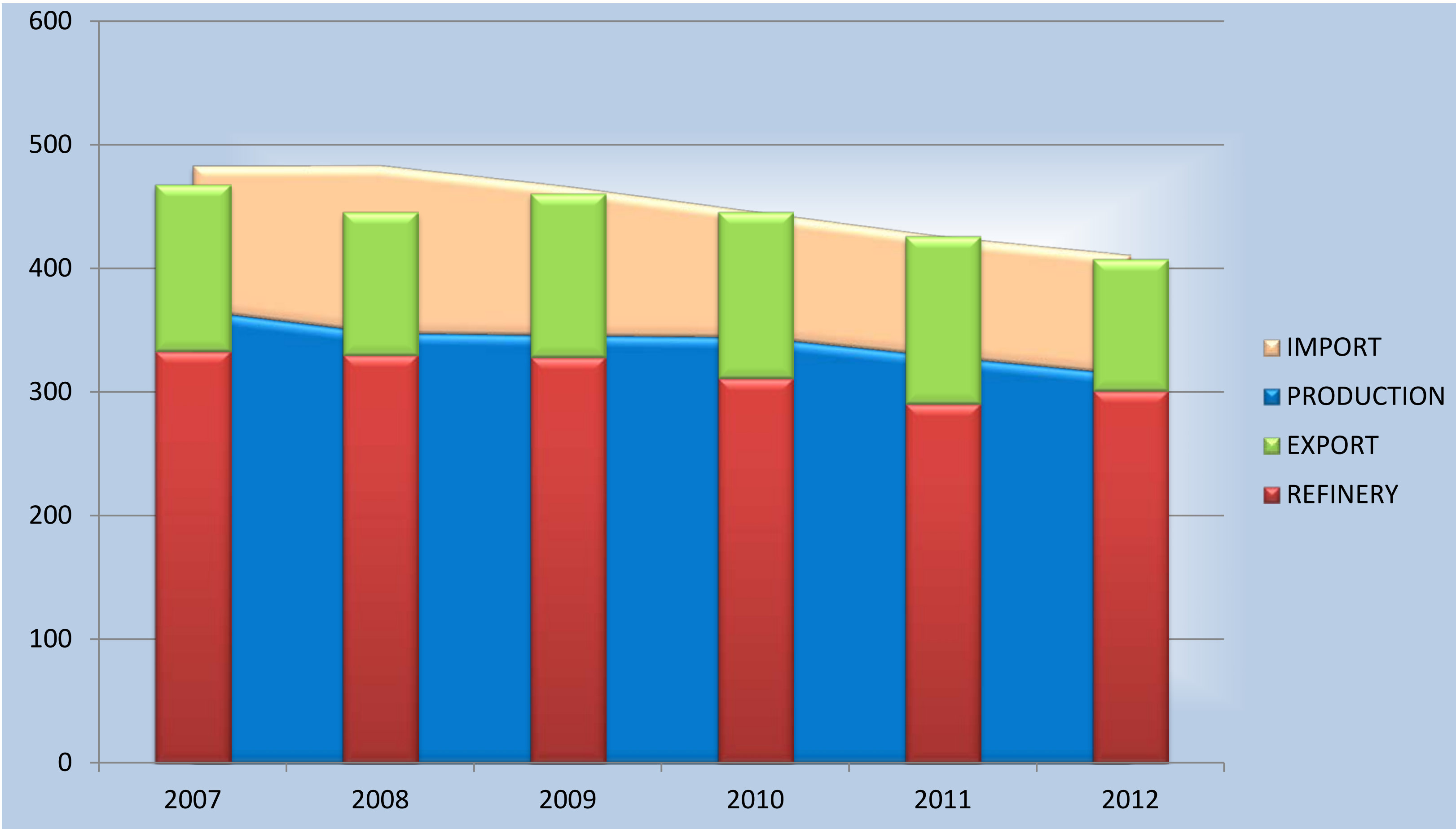
Proven	= 103.35 TSCF
Potential	= 47.35 TSCF
Total	= 150.70 TSCF

Proven	= 3,741.33 MMSTB
Potential	= 3.666.91 MMSTB
Total	= 7,408.24 MMSTB

Source: Ministry of Energy and Mineral Resources by 2013, *reprocessed by NEC*

CRUDE OIL SUPPLY

(Thousand Bbl)

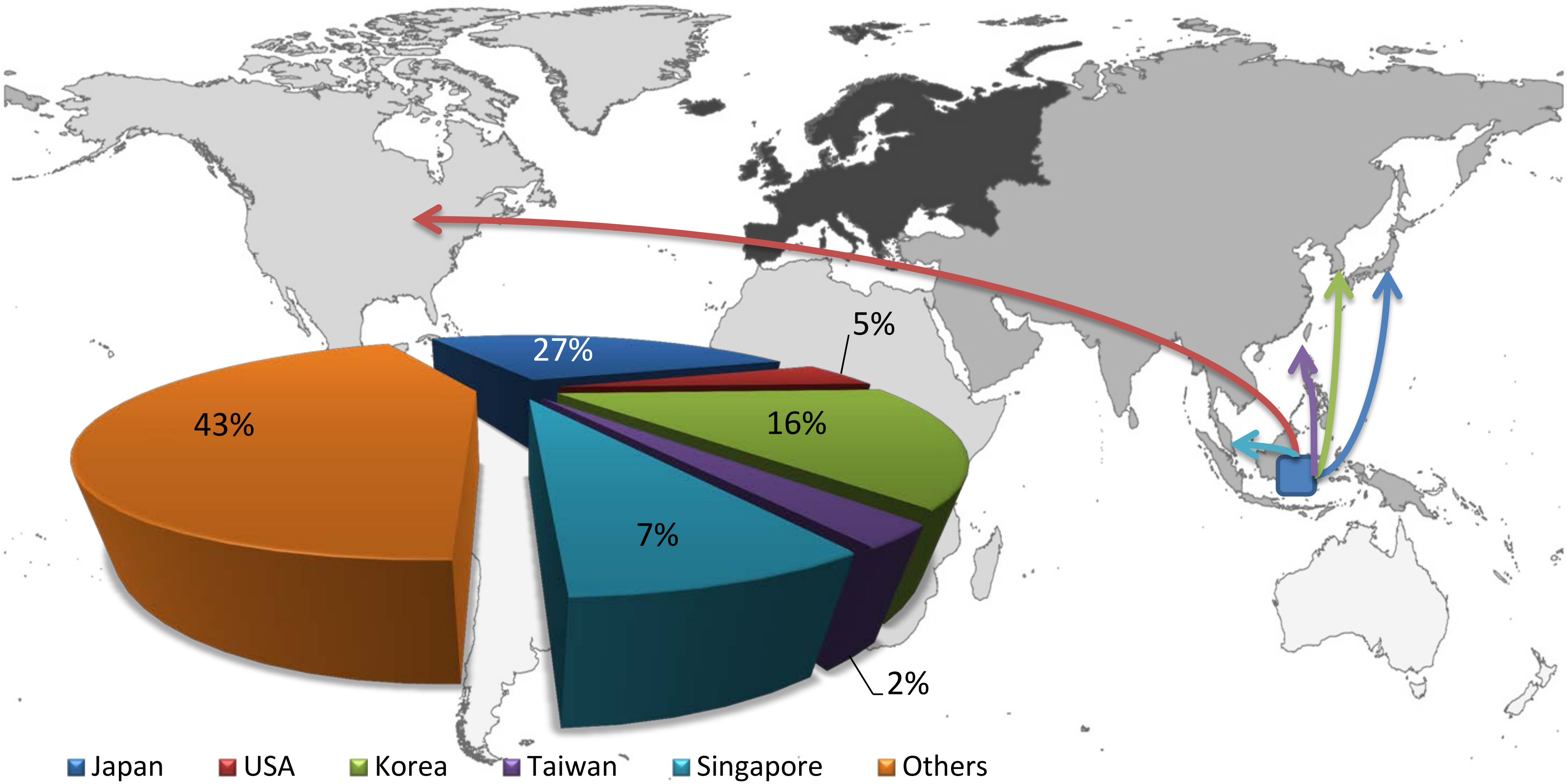


Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

Note : - discrepancy between production, export, import and domestic demand due to the existing stock

- the role of imported crude in 2012 was amounted to 32 % from the total that goes to the oil refinery

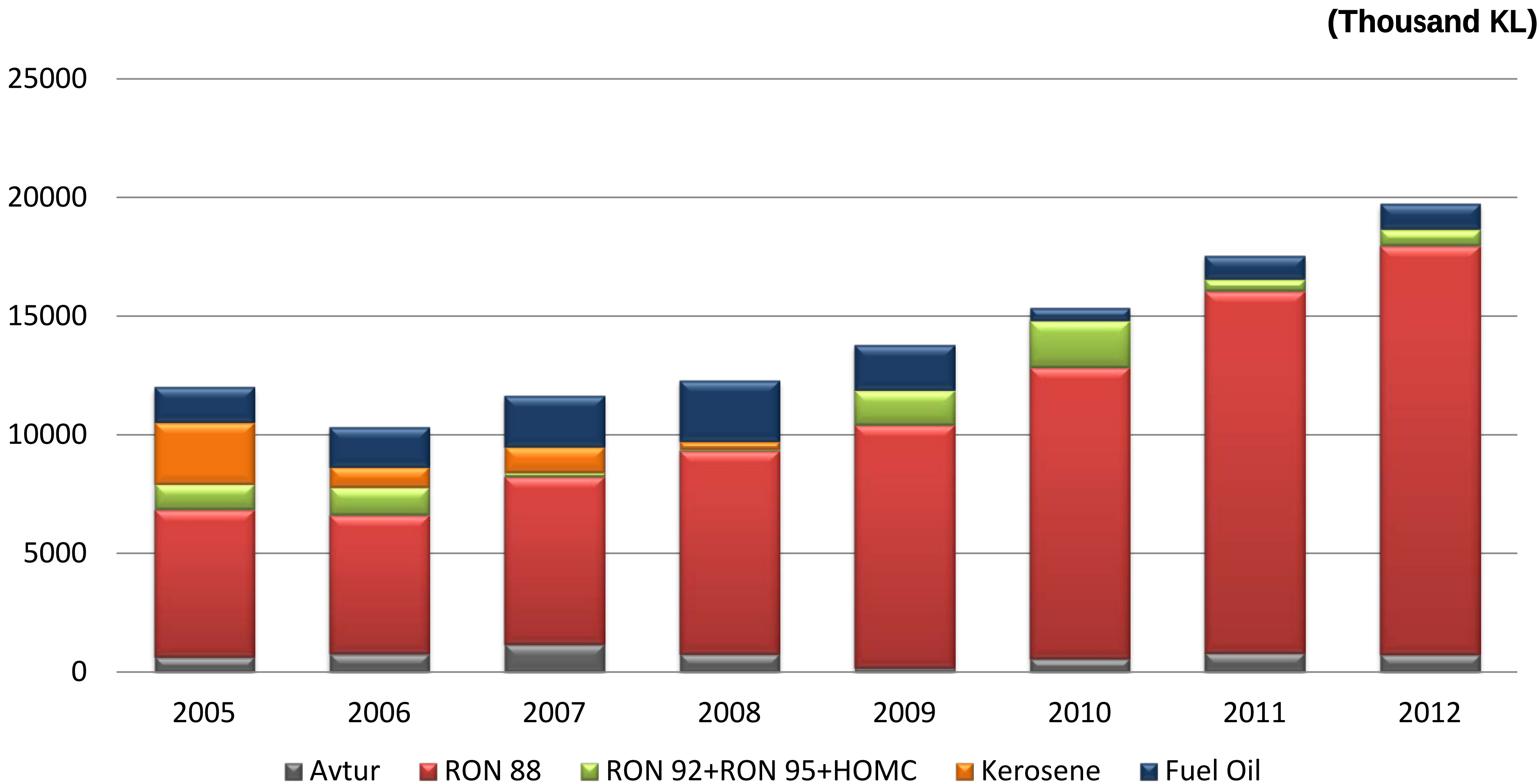
CRUDE OIL EXPORTS



By 2012, total crude oil exported from Indonesia 124.54 thousand barrels

Source: Ministry of Energy and Mineral Resources *2012, reprocessed by NEC*

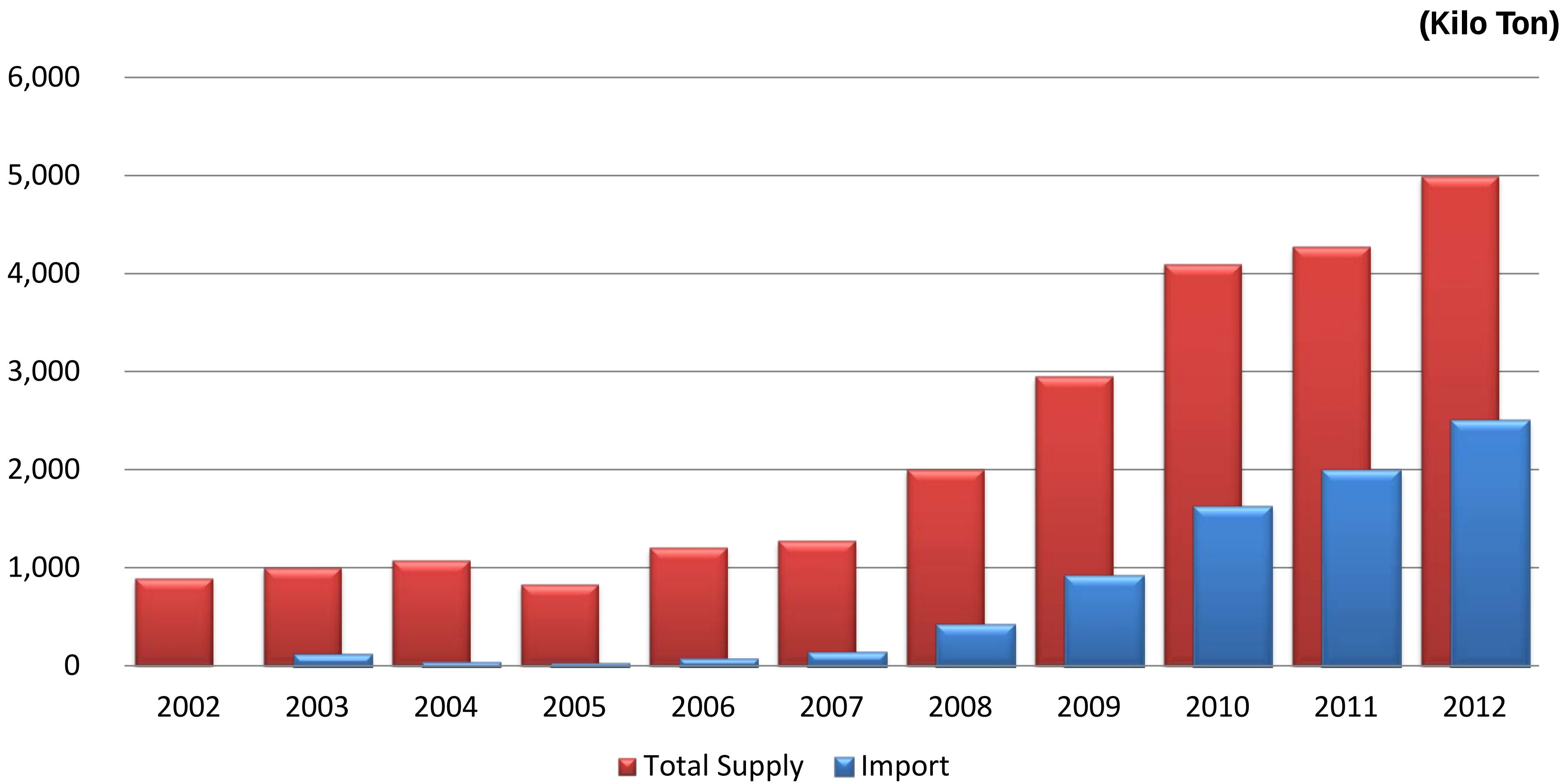
FUEL IMPORTS



Source: MEMR, reprocessed by NEC

Note : - Share of fuel imports in 2012 amounted to 54.25% (179.98 MBOE) of the total domestic fuel oil consumption
- The value of imports of RON 88 (premium) increased along with the increase in domestic demand
- Kerosene imports declined and abolished in 2009 due to LPG conversion policy

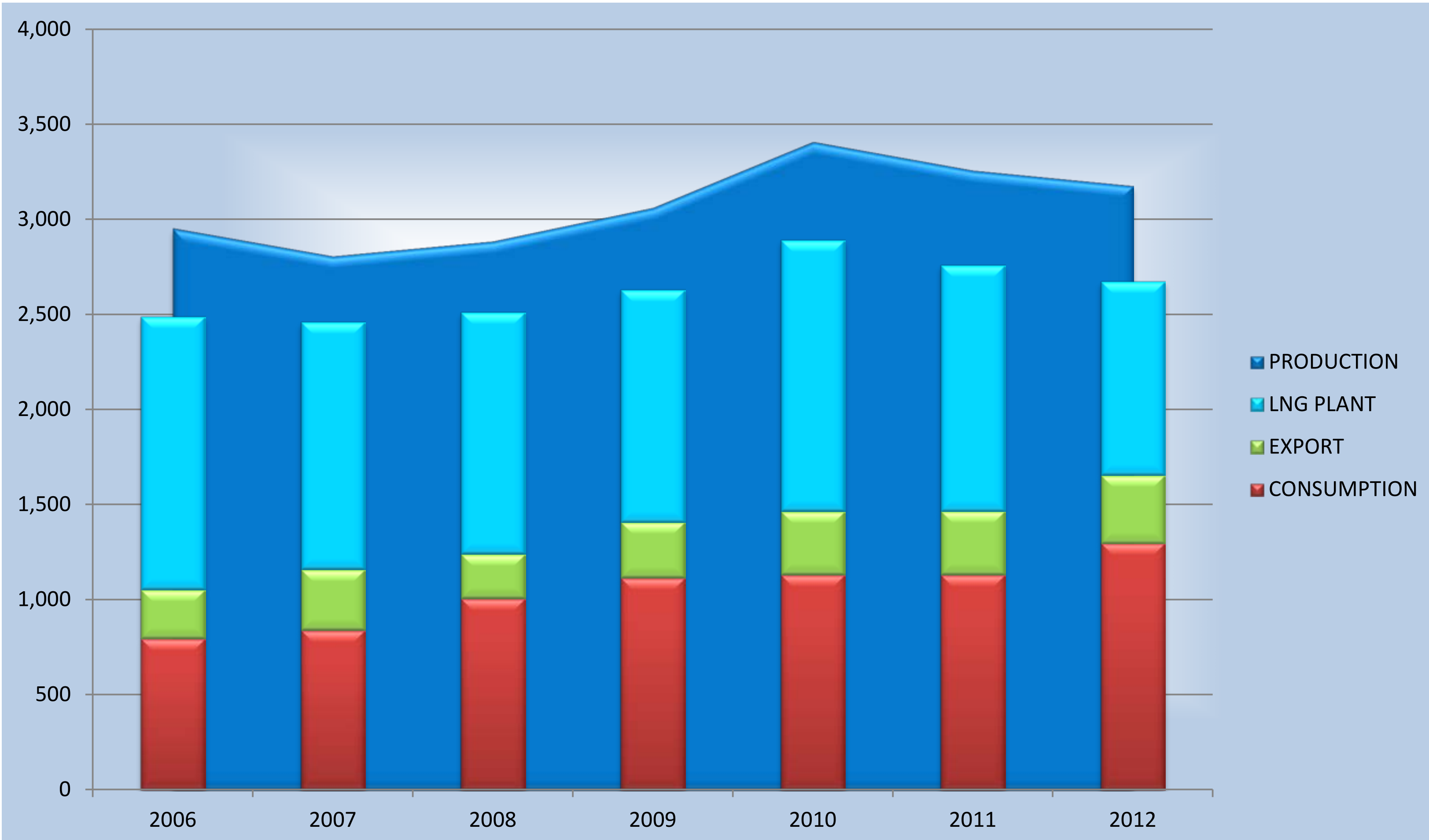
LPG IMPORTS



Source: MEMR, reprocessed by NEC
Note : - Share of LPG imports in 2012 amounted to 49.73% (21.3 MBOE) of the total LPG supply
- The value of LPG supply increased along with the increasing the demand in residential sector due to LPG conversion policy

GAS SUPPLY

(Thousand MMSCF)



Source: MEMR, reprocessed by NEC

Note : - export means export of gas through pipeline
- products of LNG refinery allocated for exports
- discrepancy between the production and the total usage of gas due to gas reinjection, own use and flare

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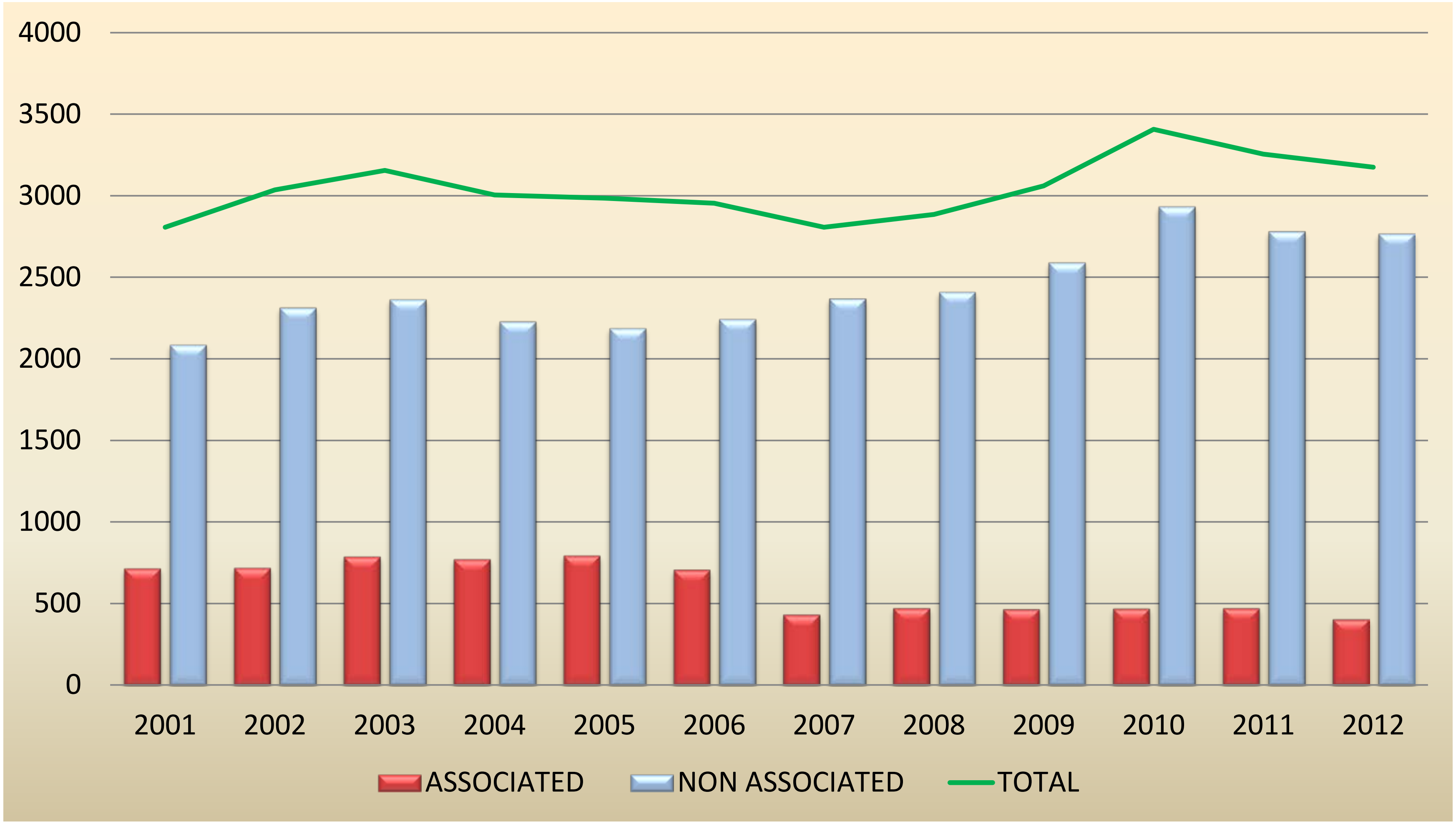
GAS PRODUCTION

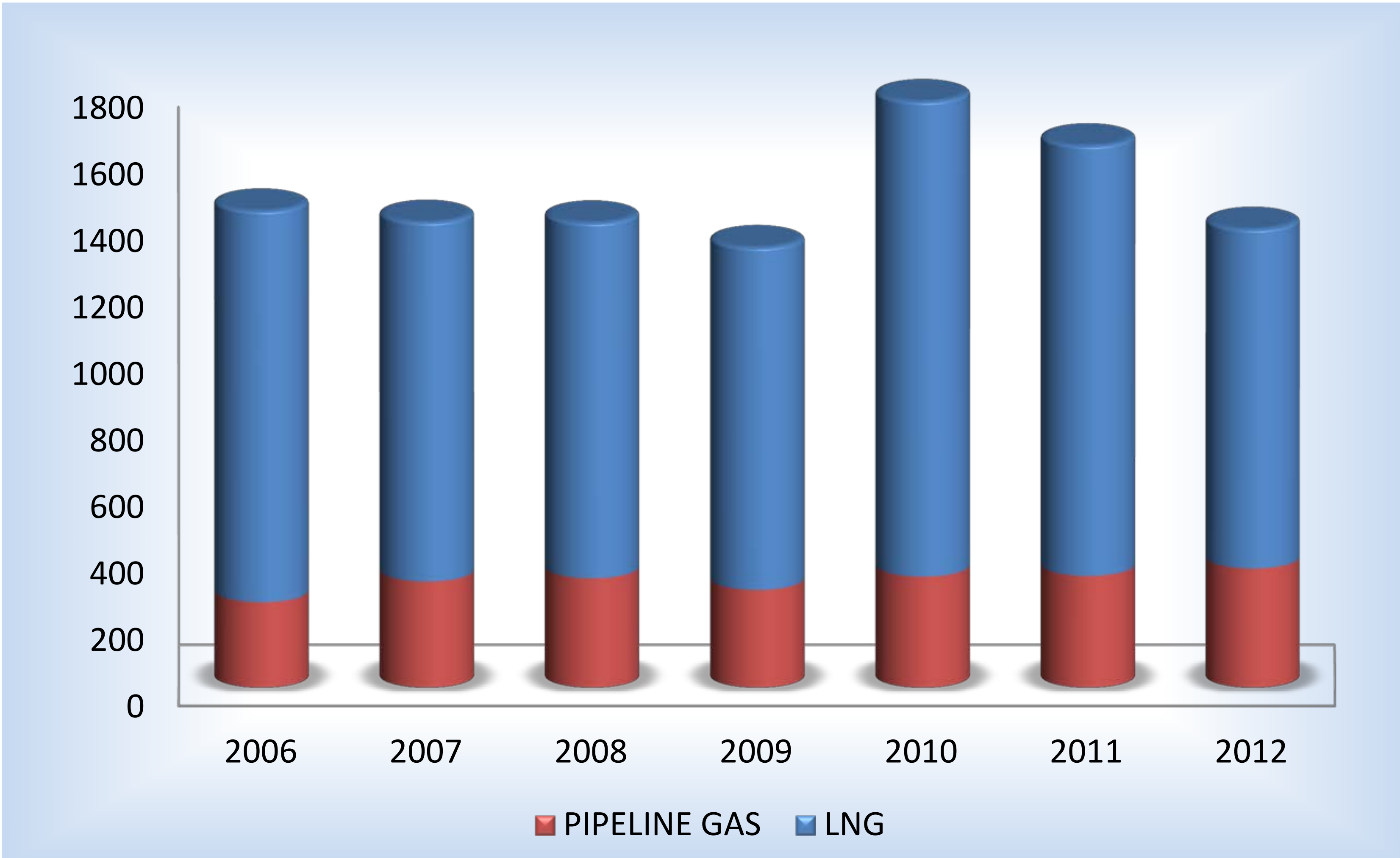
(Thousand MMSCF)



Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*
Note : associated defined as a natural gas found in association with oil, either dissolved in the oil or as a cap of free gas above the oil.

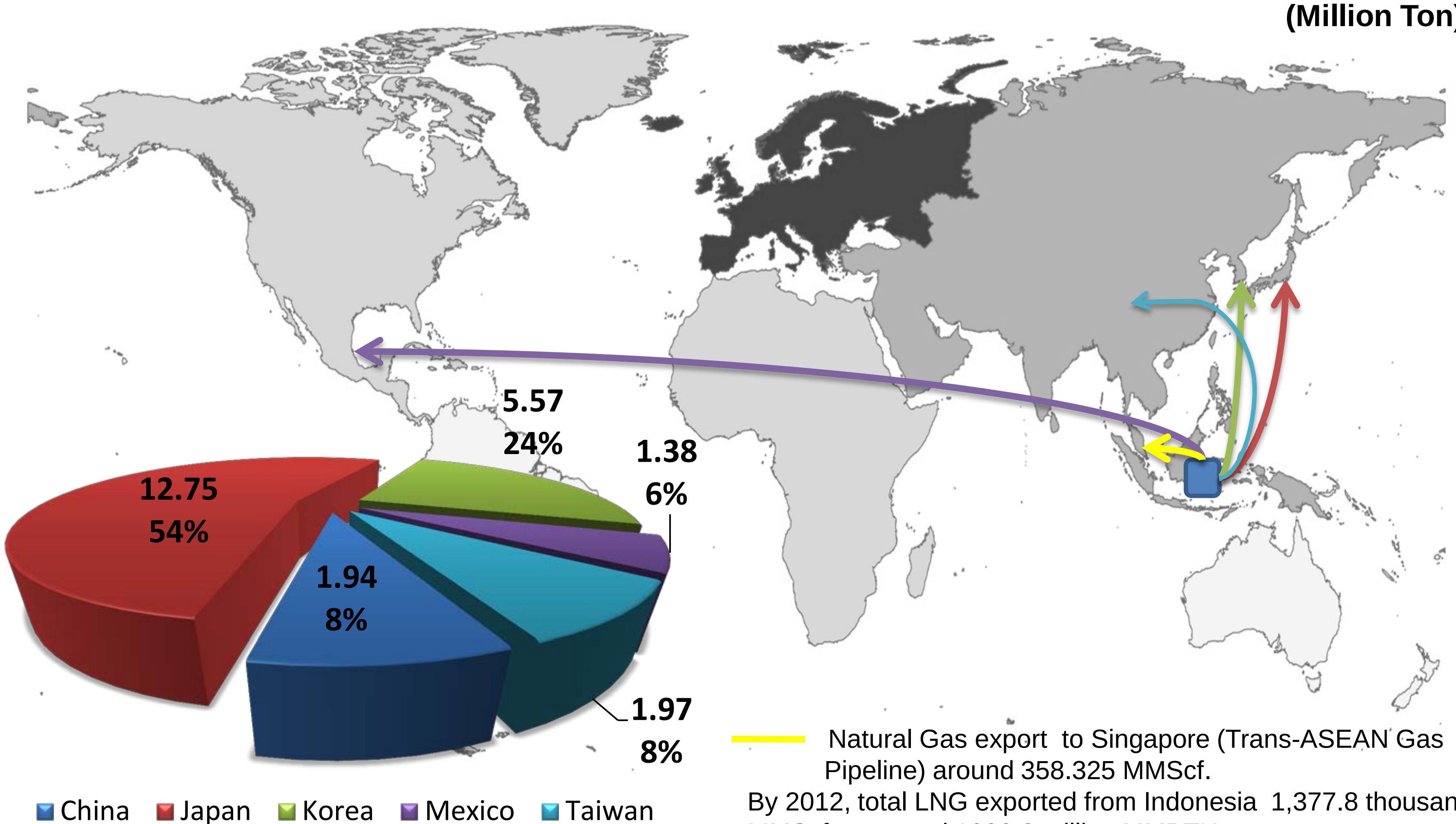
NATURAL GAS EXPORTS

(Thousand MMSCF)



Source: MEMR, reprocessed by NEC

GAS AND LNG EXPORTS



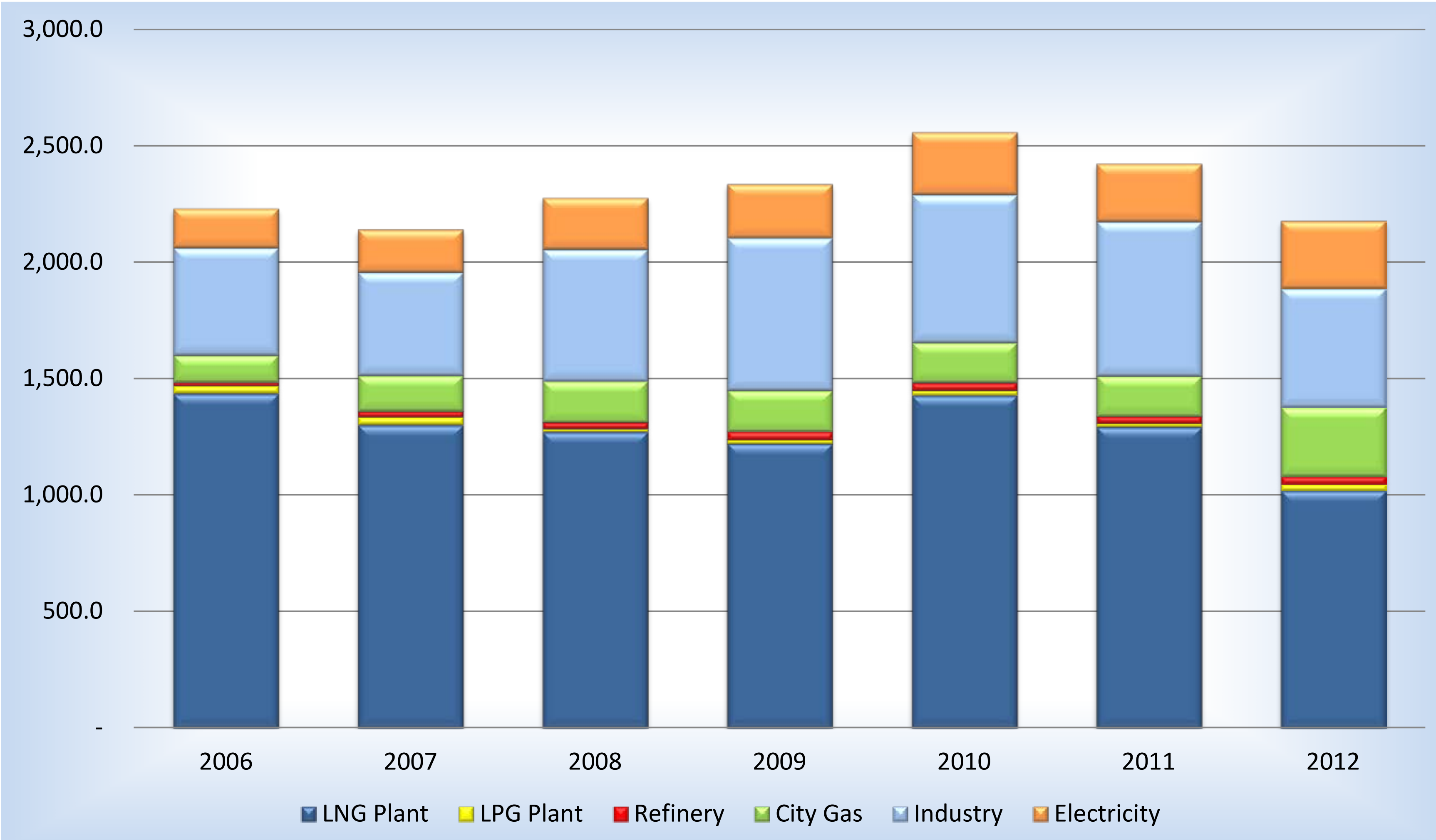
Natural Gas export to Singapore (Trans-ASEAN Gas Pipeline) around 358.325 MMScf.

By 2012, total LNG exported from Indonesia 1,377.8 thousand MMScf or around 1098.2 million MMBTU

Source: MEMR 2012, reprocessed by NEC

NATURAL GAS CONSUMPTION

(Thousand MMSCF)



Source: MEMR, reprocessed by NEC

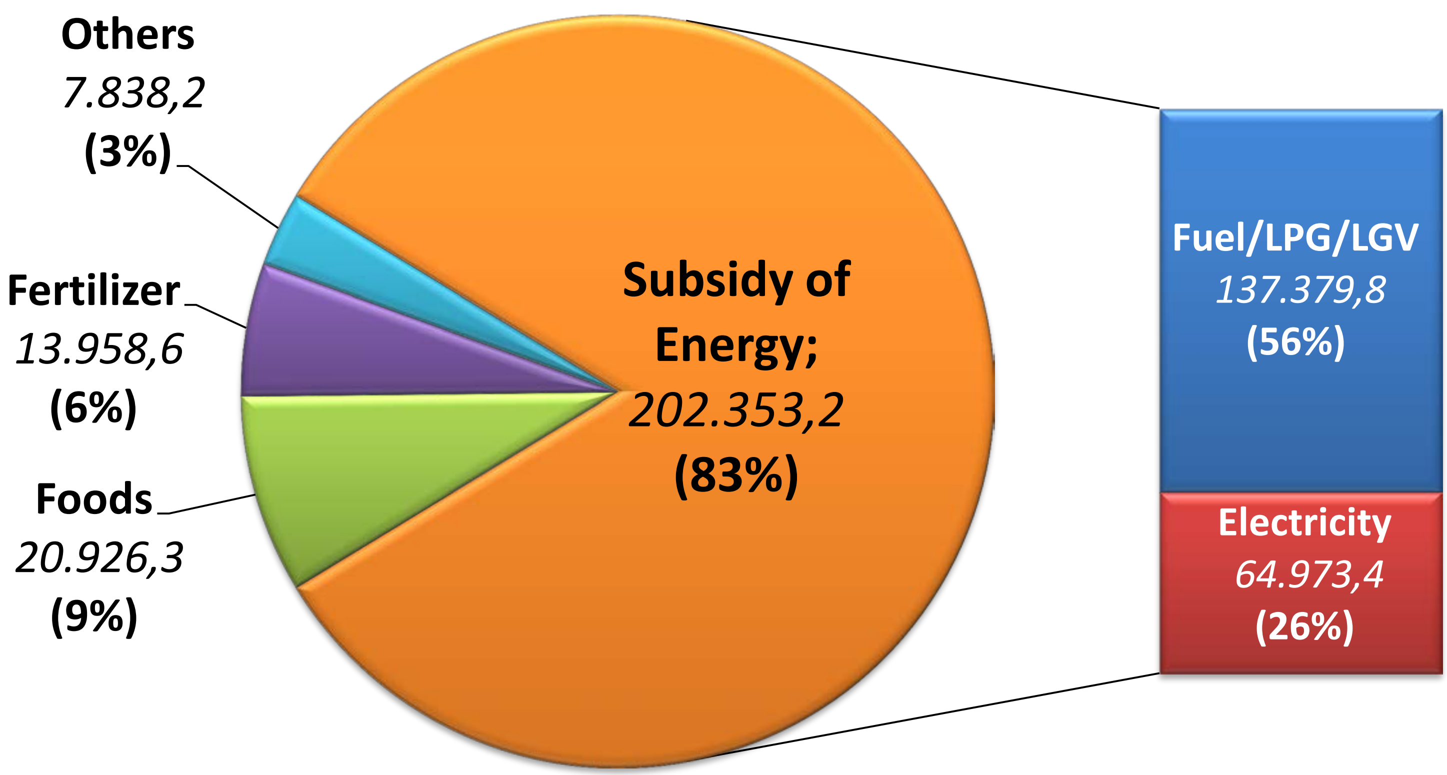
FUEL OIL SUBSIDY

(Trillion IDR)

Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*
Note : rise in oil prices due to geopolitical crisis in the Middle East and Libya

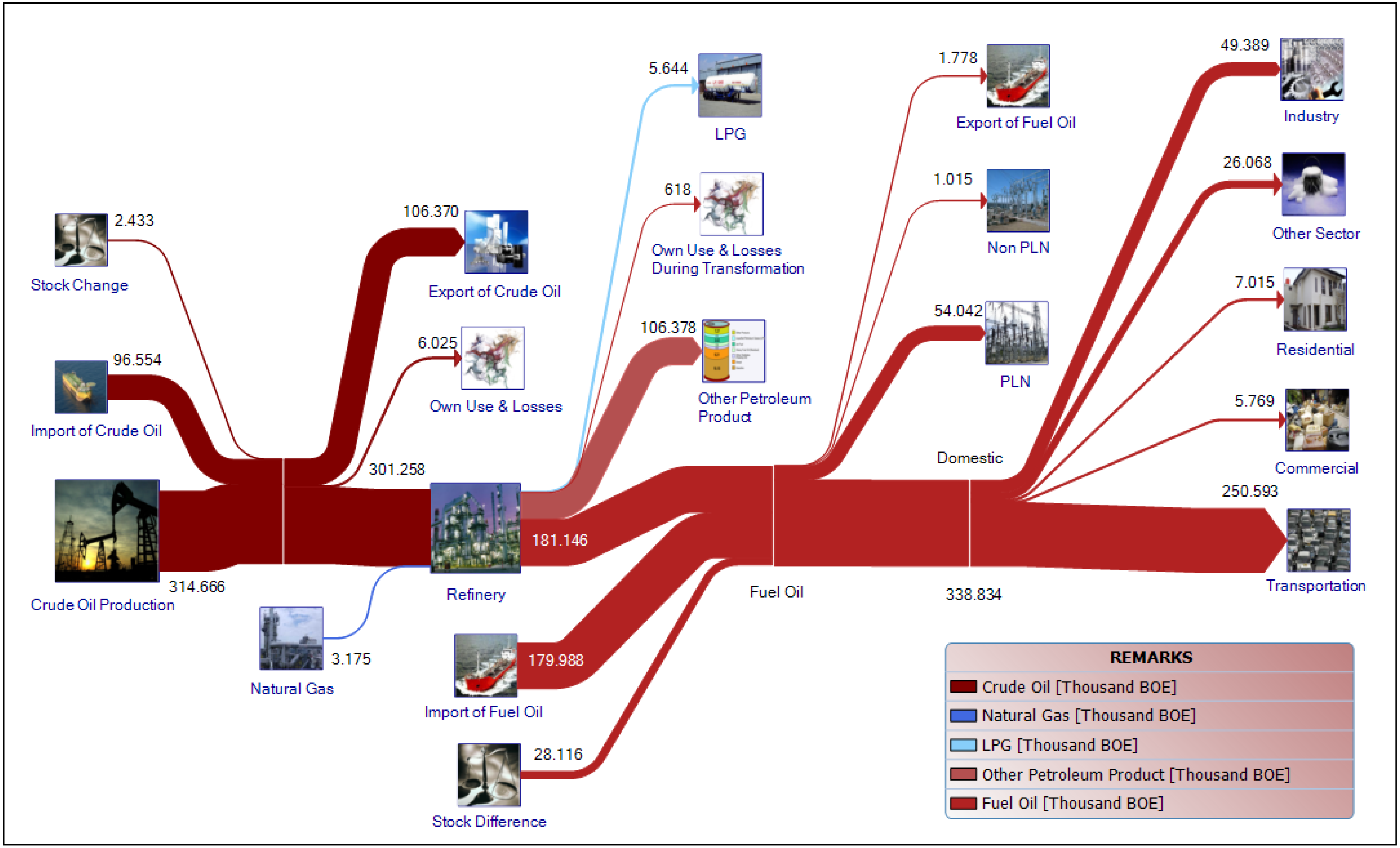
SHARES OF TOTAL ENERGY SUBSIDY, 2012

(Trilion Rupiahs)



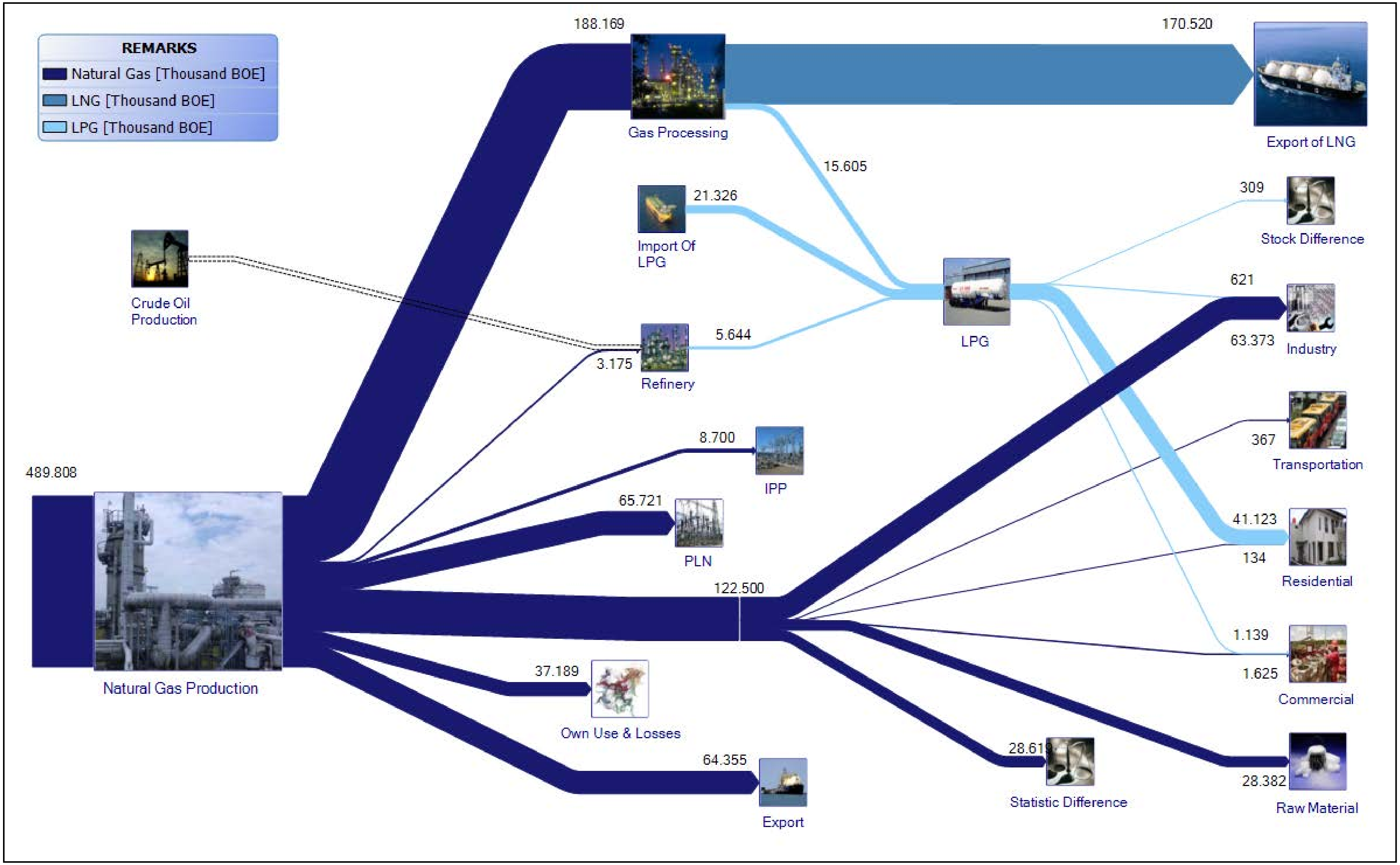
Source: Ministry of Finance, *reprocessed by NEC*

OIL FLOW IN 2012



Source: National Energy Council

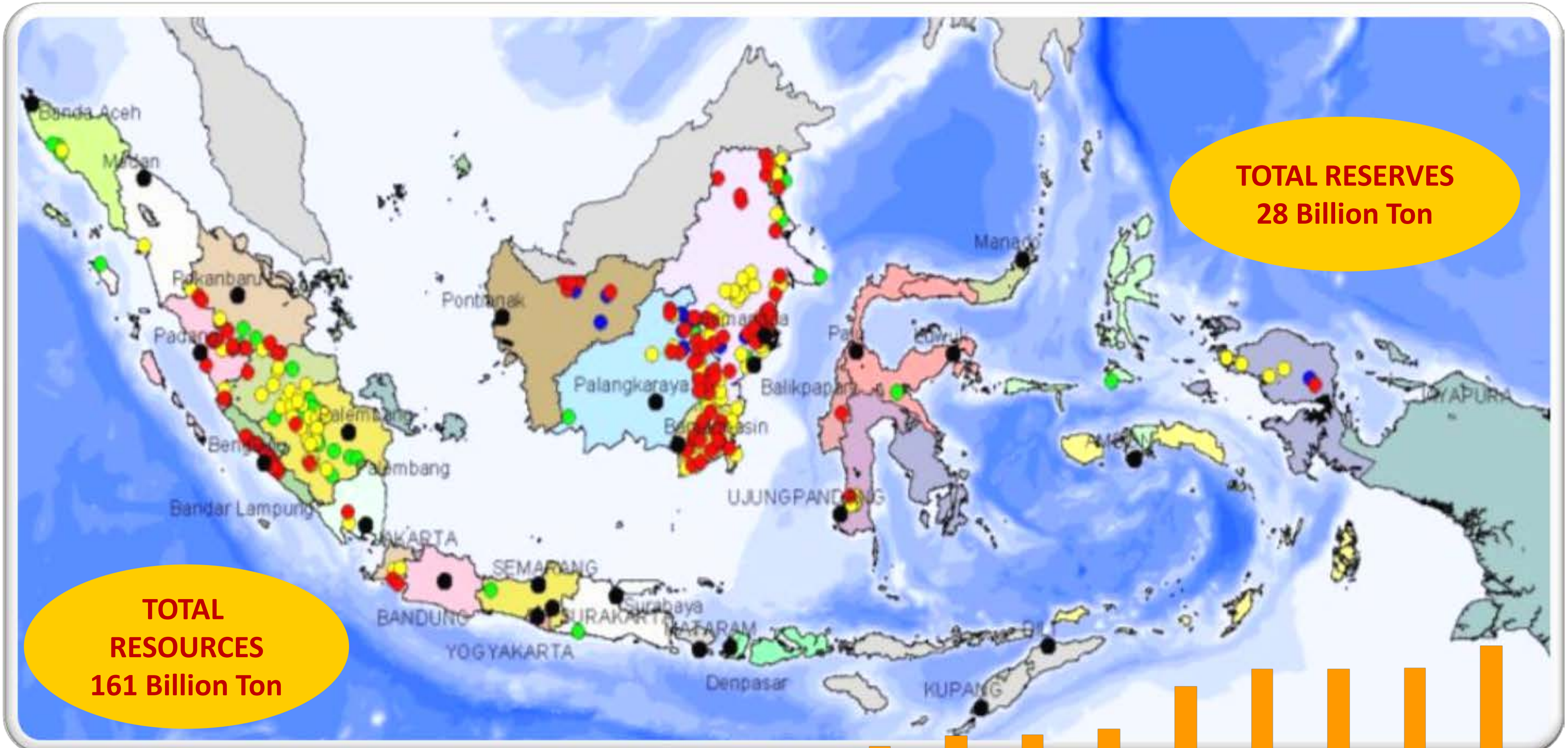
GAS FLOW IN 2012



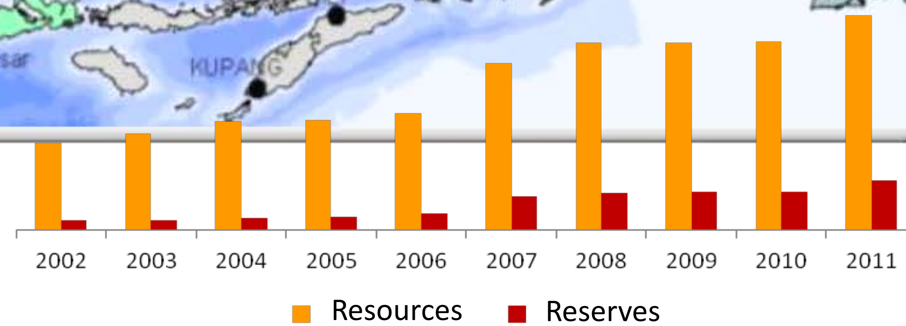
Source: National Energy Council

COAL

COAL RESOURCES AND RESERVES



Data by 2012

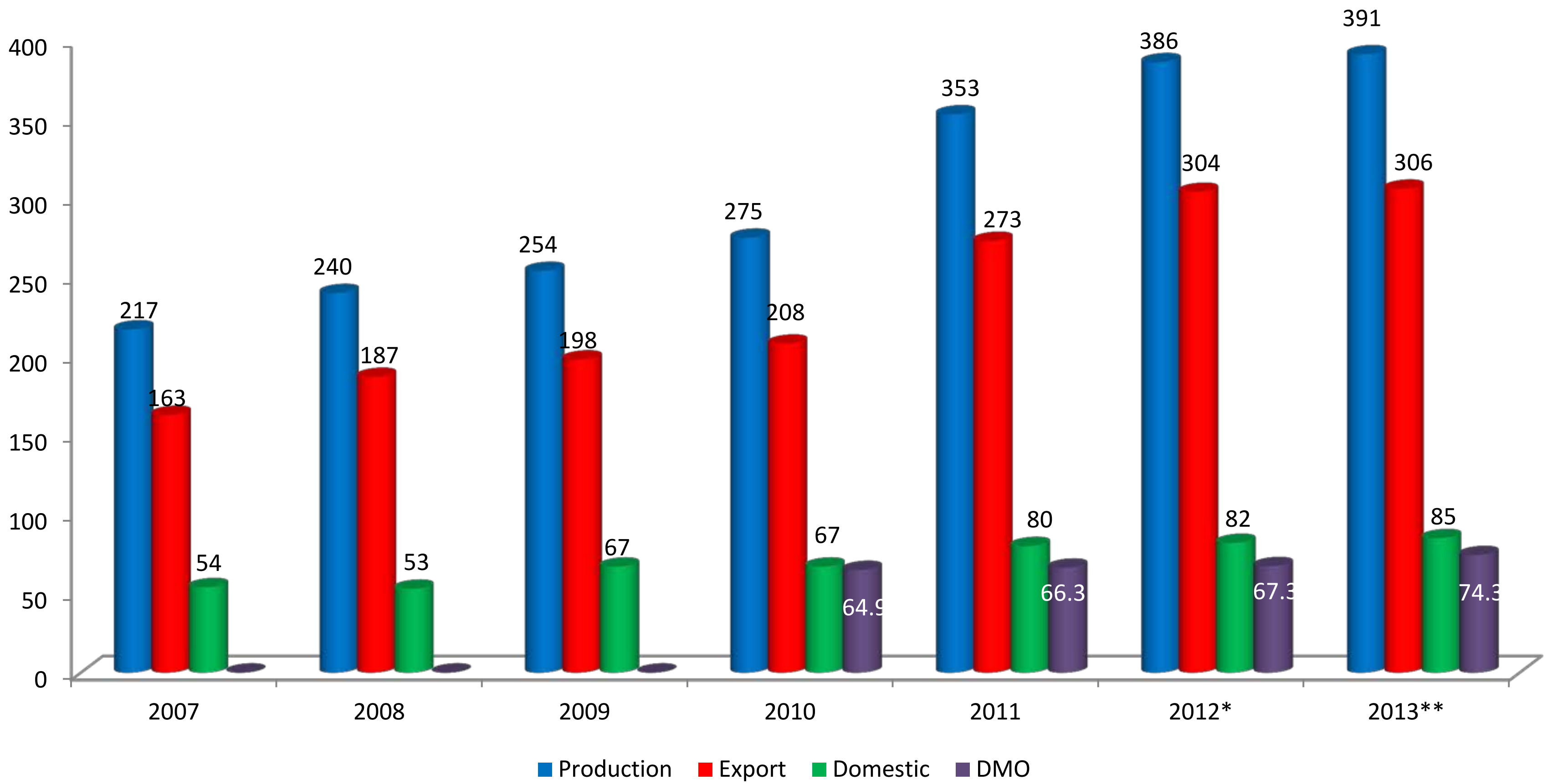


Very High	(> 7.100 kal/gr)
High	(6.100 – 7.100 kal/gr)
Medium	(5.100 – 6.100 kal/gr)
Low	(< 5.100 kal/gr)

Source: Ministry of Energy and Mineral Resources

COAL UTILIZATION

(Million Ton)



Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

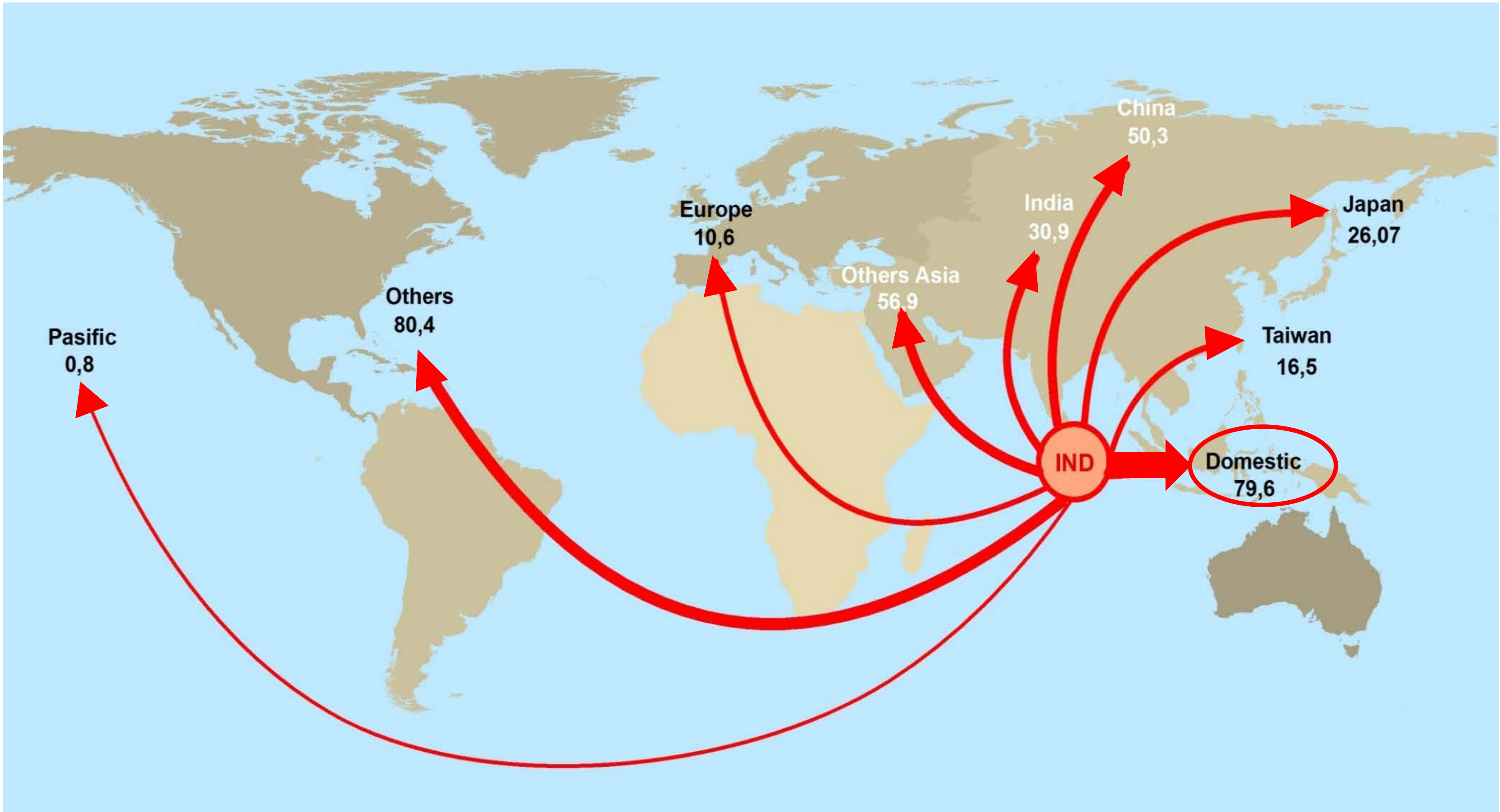
Note : *) Realization forecast in year 2012

Coal's Domestic Market Obligation (DMO) started to be enforced in 2010

*)) 2013 Target

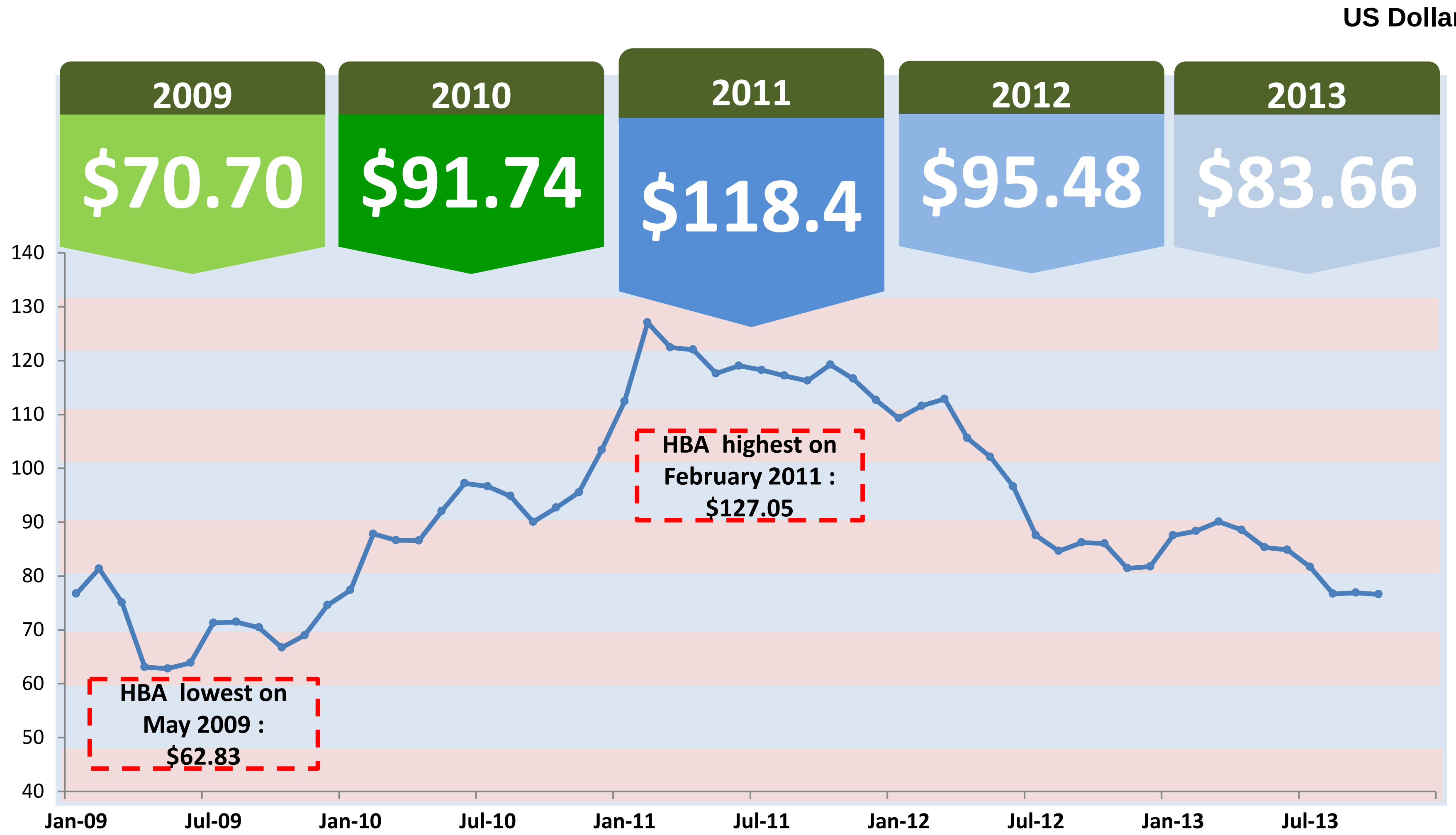
COAL EXPORT BY COUNTRY DESTINATION

(Million Ton)



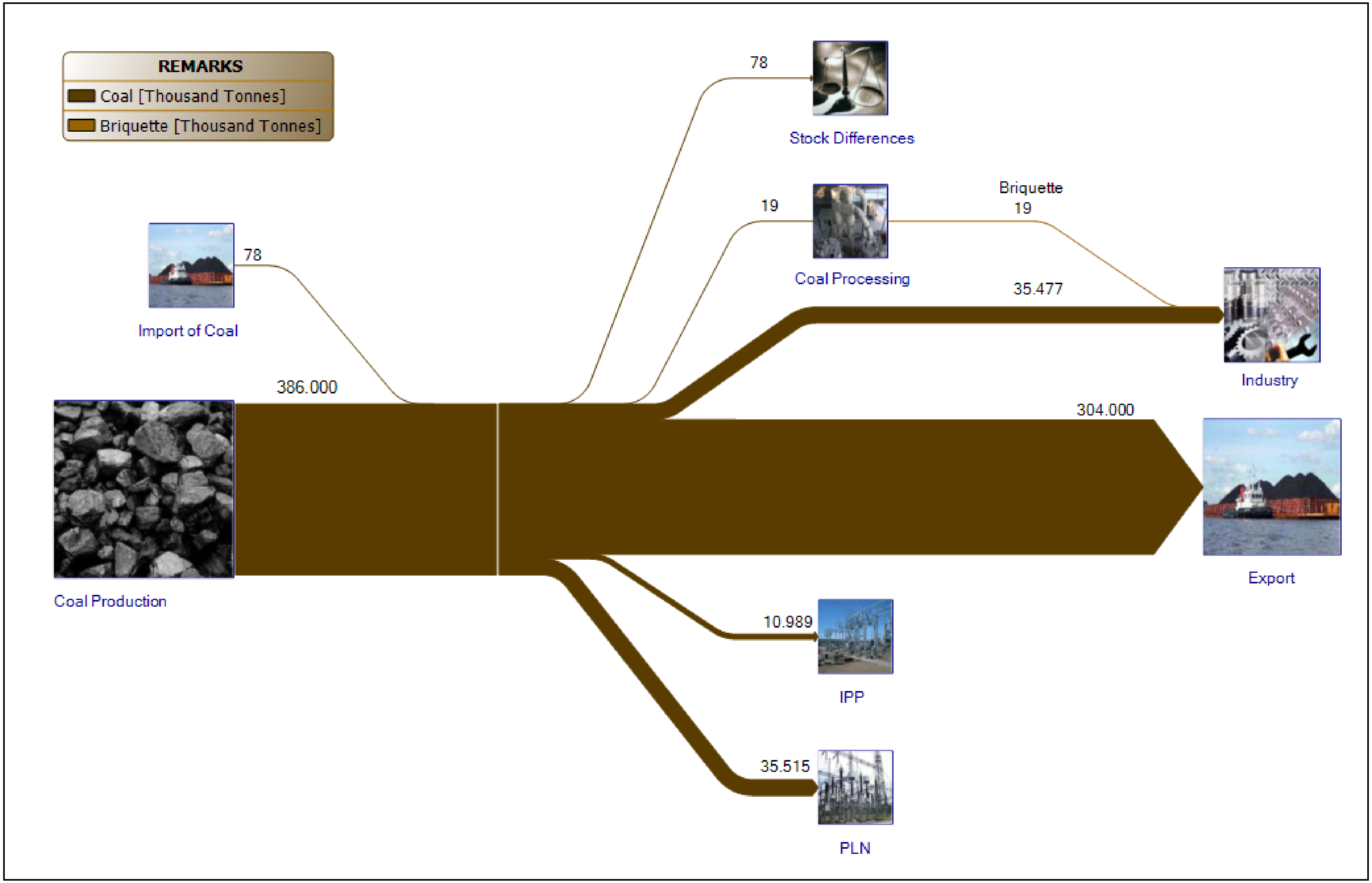
Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

COAL REFERENCE PRICE (HBA)



(Data source: MEMR, reprocessed by NEC)

COAL FLOW



Source: National Energy Council, 2012

ELECTRICITY

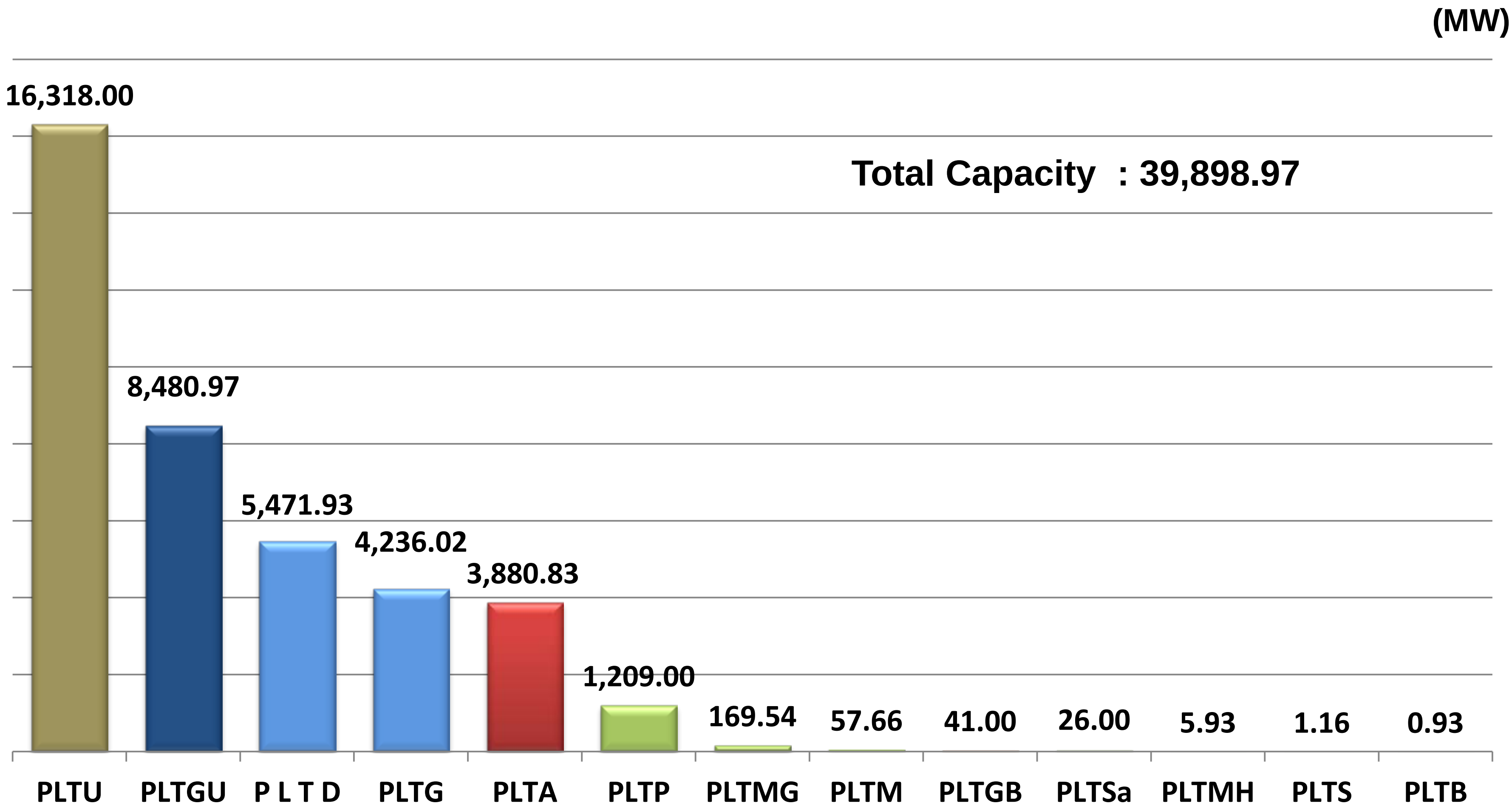
ELECTRICITY POWER CONDITION

No	DESCRIPTION	UNIT	2011	2012		2013
			Realization	Plan	Realization	Plan
1	Demand Growth of Electricity	%	10,1	8,4	8,4	8,6
2	Electrification Ratio	%	72,95	75,3	76,6	79,3
3	Village Electrification Ratio	%	96	96,7	96,7	97,8
4	Total Installed Capacity	MW	39.885	44.224	44.124	48.161
	a. PLN	MW	30.529	33.454	32.108	35.564
	b. *Independent Power Producer* (IPP)	MW	7.653	9.066	10.287	10.718
	c. *Private Power Utilities* (PPU)	MW	1.704	1.704	1.729	1.729
5	PLN Electricity Production and Purchase of Electricity	GWh	175.213	190.940	193.663	207.409
6	Rural Electricity					
	a. Sub Station Distribution	MVA	368	213,1	249	216,8
	b. Distribution Network	KMs	17.570,7	8.590,3	11.311,5	9.244,3
	c. Cheap and EfficientEelectricity	RTS*)	-	83.000	60.702	95.227

Source : Ministry of Energy and Mineral Resources & *National Electric Company* (PLN)

*) RTS = Target Households (<450 Watt)

POWER PLANT CAPACITY, 2011

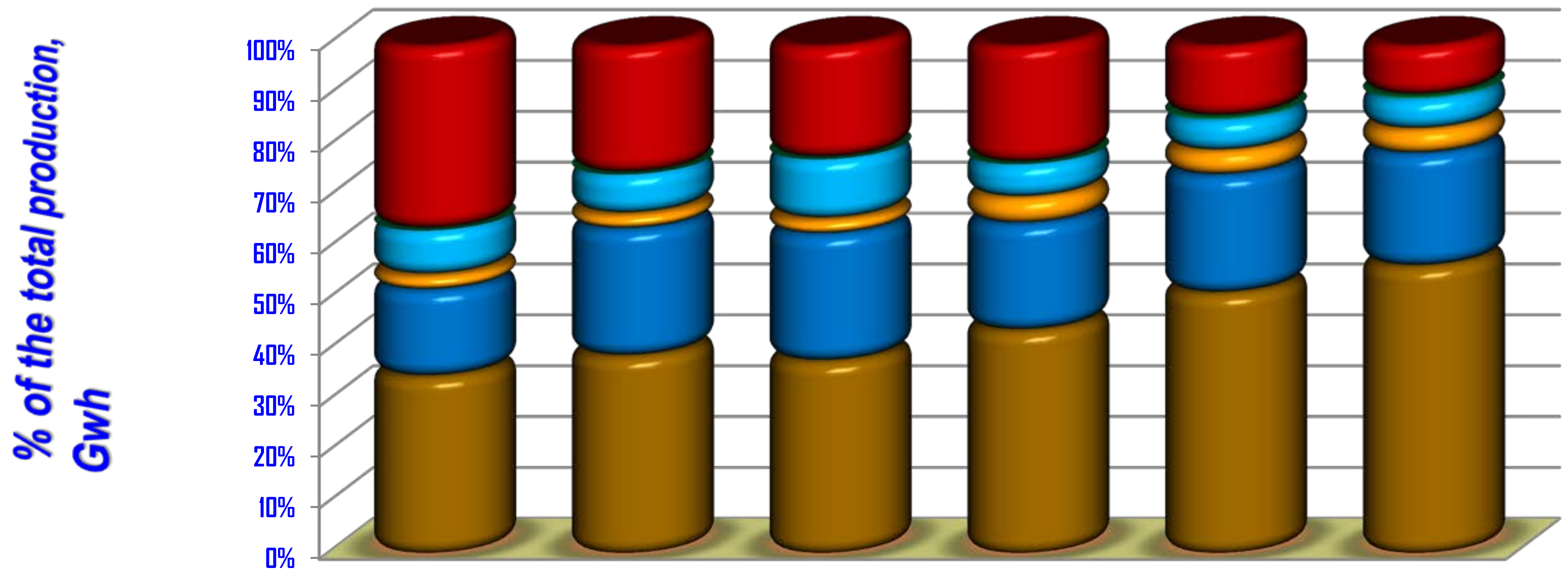


Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

Note : the figures on the PLTU (steamed power plant) does not only use coal as the primary fuel, but there is also the use of gas, the MFO and HSD.

TARGET AND REALISATION OF THE ENERGY MIX POWER PLANT (2008 – 2013)

(%)

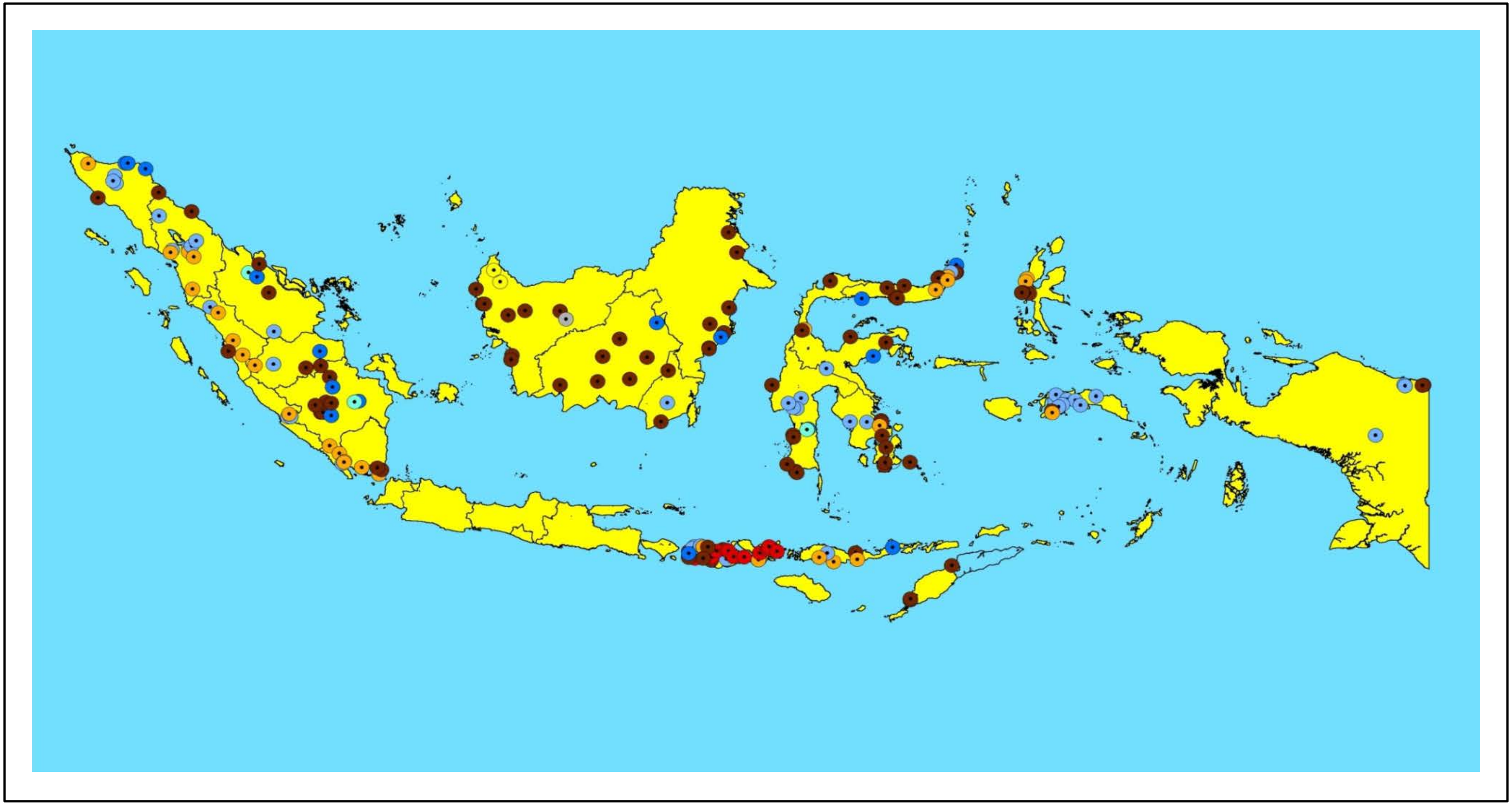


	2008	2009	2010	2011	2012	2013*)
Fuel OIl	36%	25%	22%	22,95%	15,02%	9,70%
Bio-Diesel & Other NRE	0%	0%	0%	0,07%	0,12%	0,52%
Hydropower	9%	8%	12%	6,80%	6,44%	6,19%
Geothermal	3%	3%	3%	5,13%	4,75%	4,80%
Gas	17%	25%	25%	21,00%	22,60%	22,12%
Coal	35%	39%	38%	44,06%	51,08%	56,66%

**) Temporary Data*

Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

POWER PLANTS LOCATION



Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

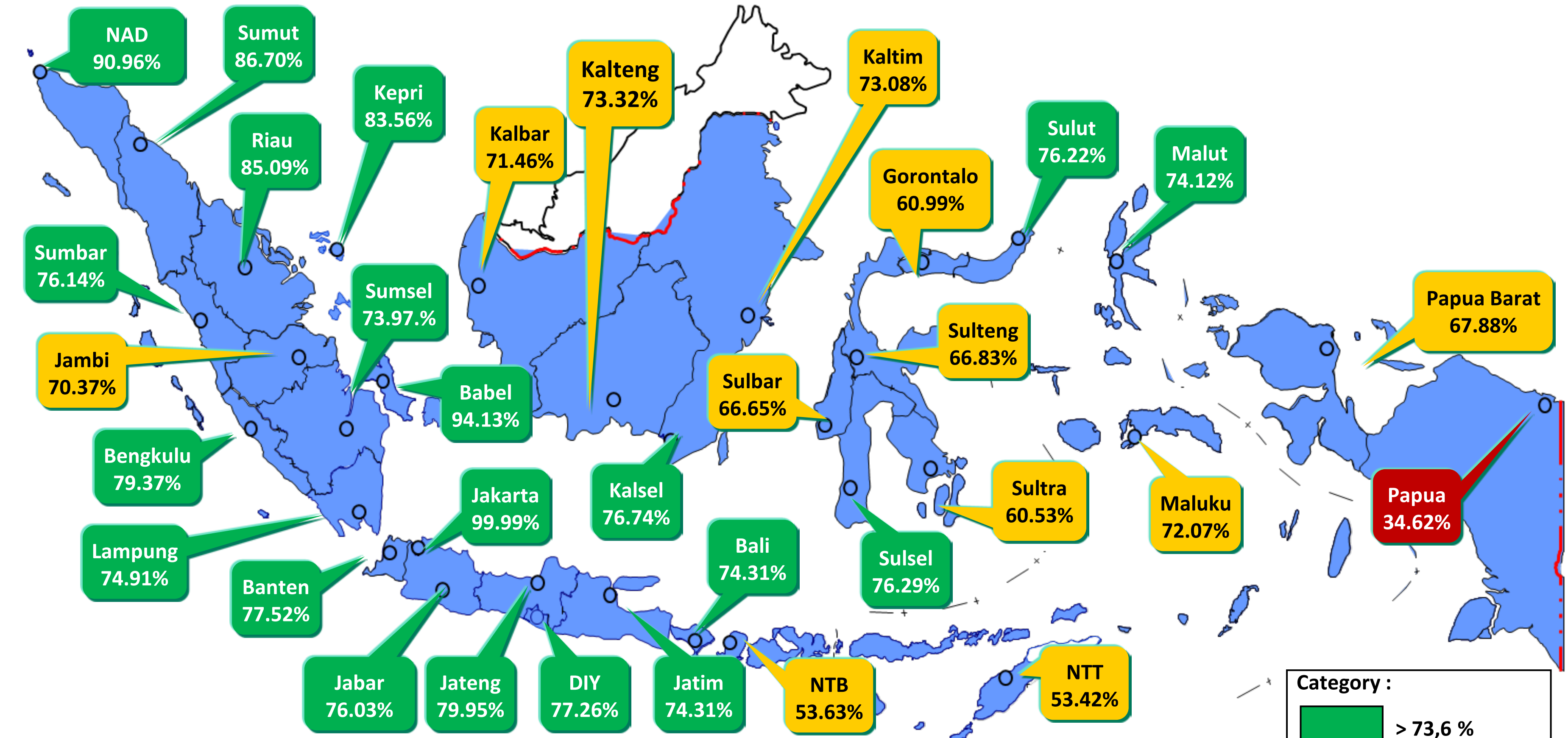
PLTA Plan
PLGB Plan
PLM Plan
PLTU Plan
PLTG PLan
PLGU Plan
PLTP Plan
PLTD Plan

ELECTRICITY TRANSMISSION NETWORK

Source: Source: Ministry of Energy and Mineral Resources (EBTKE,2012) *, reprocessed by NEC*

Existing Transmission Line ▪▪▪▪▪ Transmission Line Plan

ELECTRIFICATION RATIO 2012



	REALISATION (%)						PLAN (%)	
	2007	2008	2009	2010	2011	2012	2013	2014
Electrification Ratio	64.3%	65.1%	65.8%	67.2%	73.0%	76.6%	79.3%	81.4%
RPJM Target				67.2%	70.4%	73.6%	76.8%	80.0%

Category :

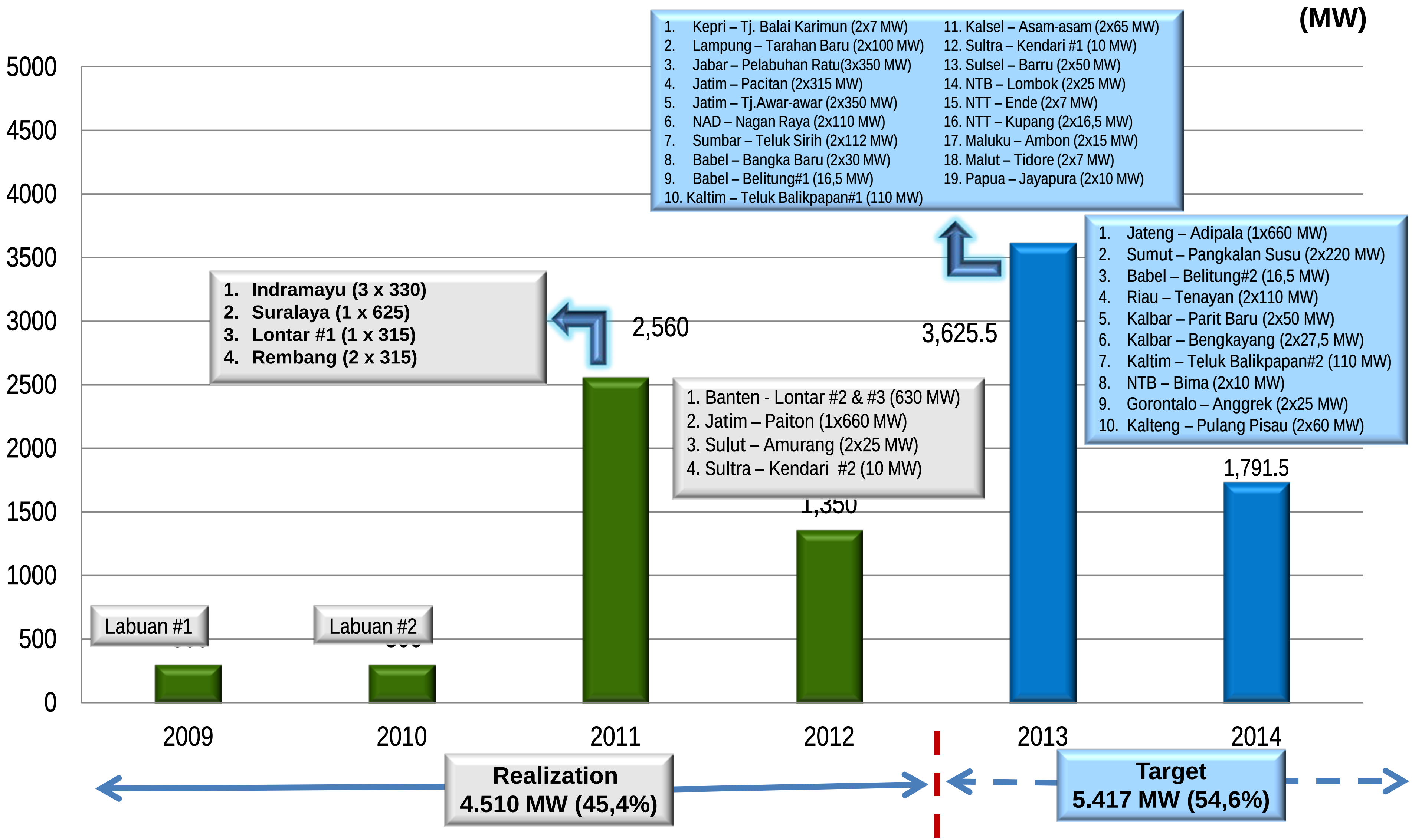
- > 73,6 %
- 50 – 73,6 %
- 20 - 50 %

Source: Source: Ministry of Energy and Mineral Resources (EBTKE,2012) , *reprocessed by NEC*

The category to be attuned to the RPJM Target 2012 (C%) ; RPJM = Long and Medium Term Development PLan

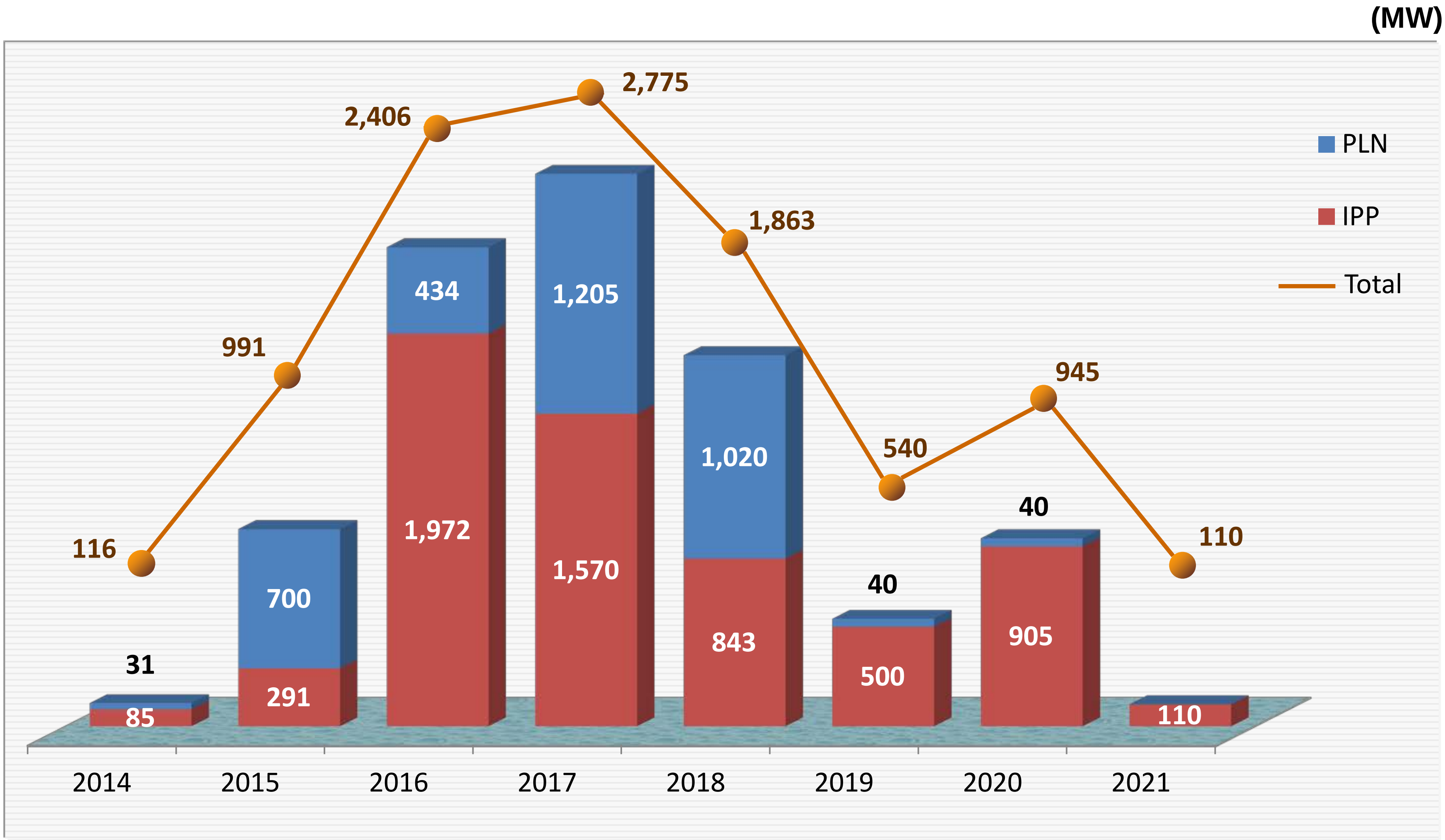
PROGRESS of FTP-I and FTP-II

10,000 MW FAST TRACK PROGRAM PHASE I (FTP I)



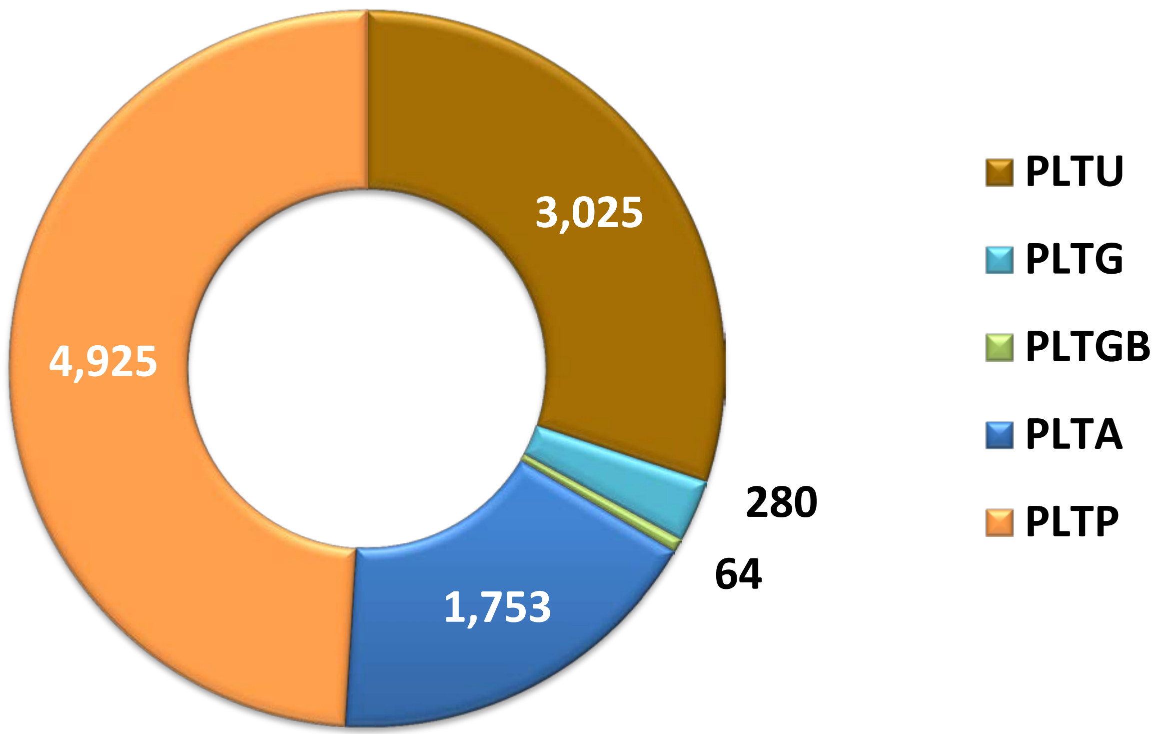
Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

10,000 MW FAST TRACK PROGRAM PHASE II (FTP II)



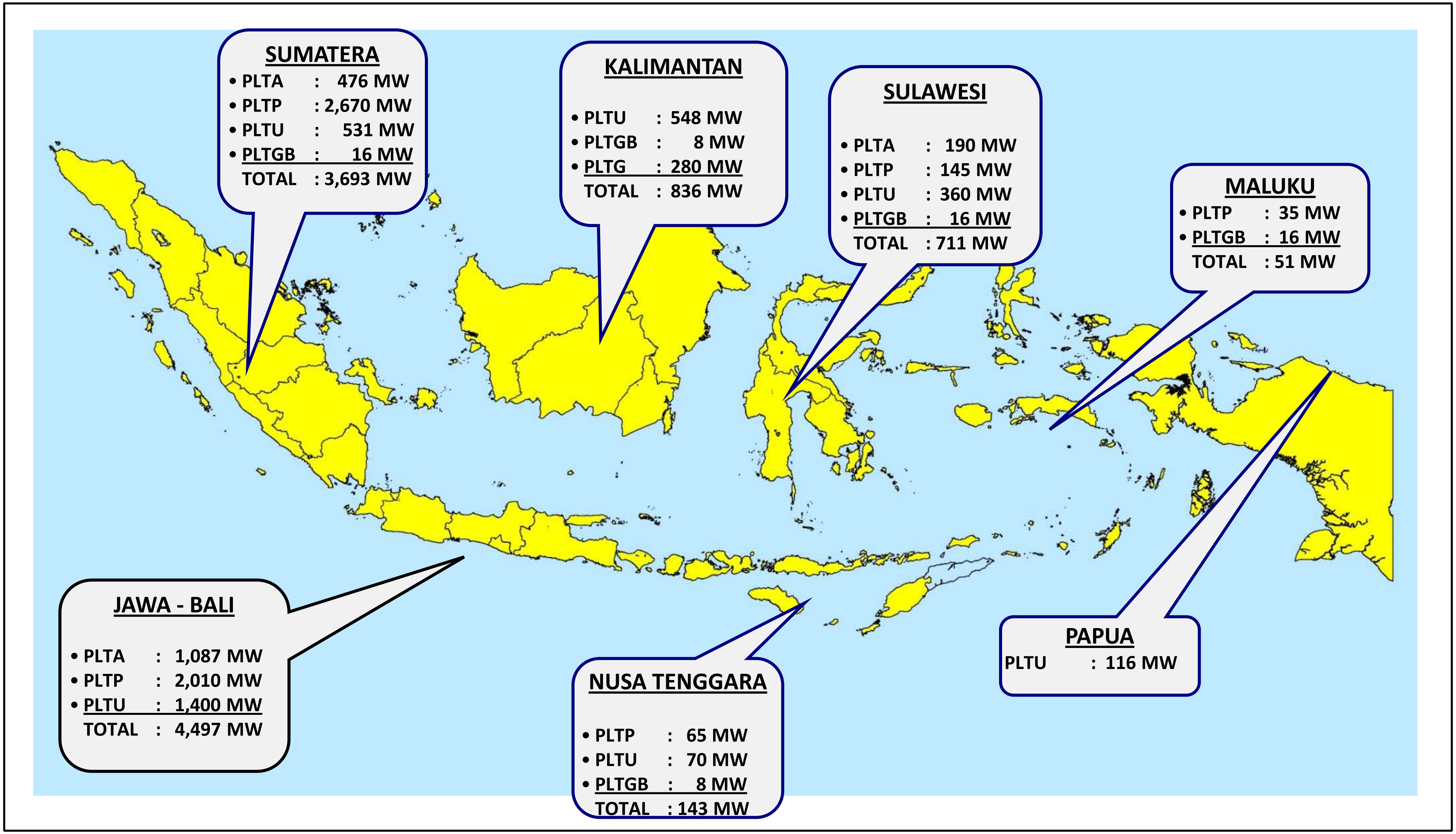
Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

COMPOSITION OF 10,000 MW FAST TRACK PROGRAM PHASE II



Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

DISTRIBUTION of 10,000 MW POWER PLANT PHASE II

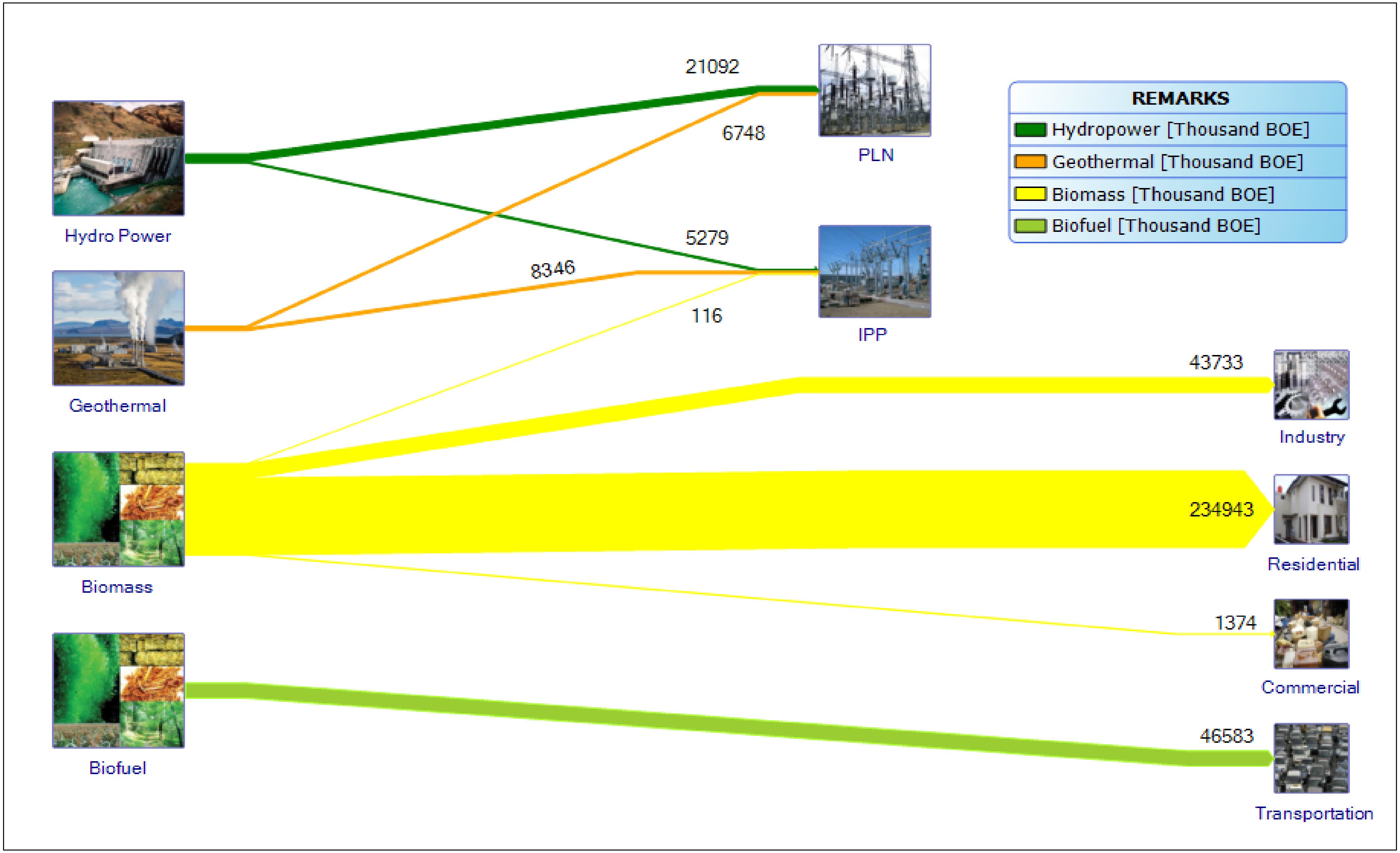


Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

*) EMR Ministerial Regulation Number 01 Year 2012

NEW AND RENEWABLE ENERGY

NEW AND RENEWABLE ENERGY FLOW



Source: NEC, 2012

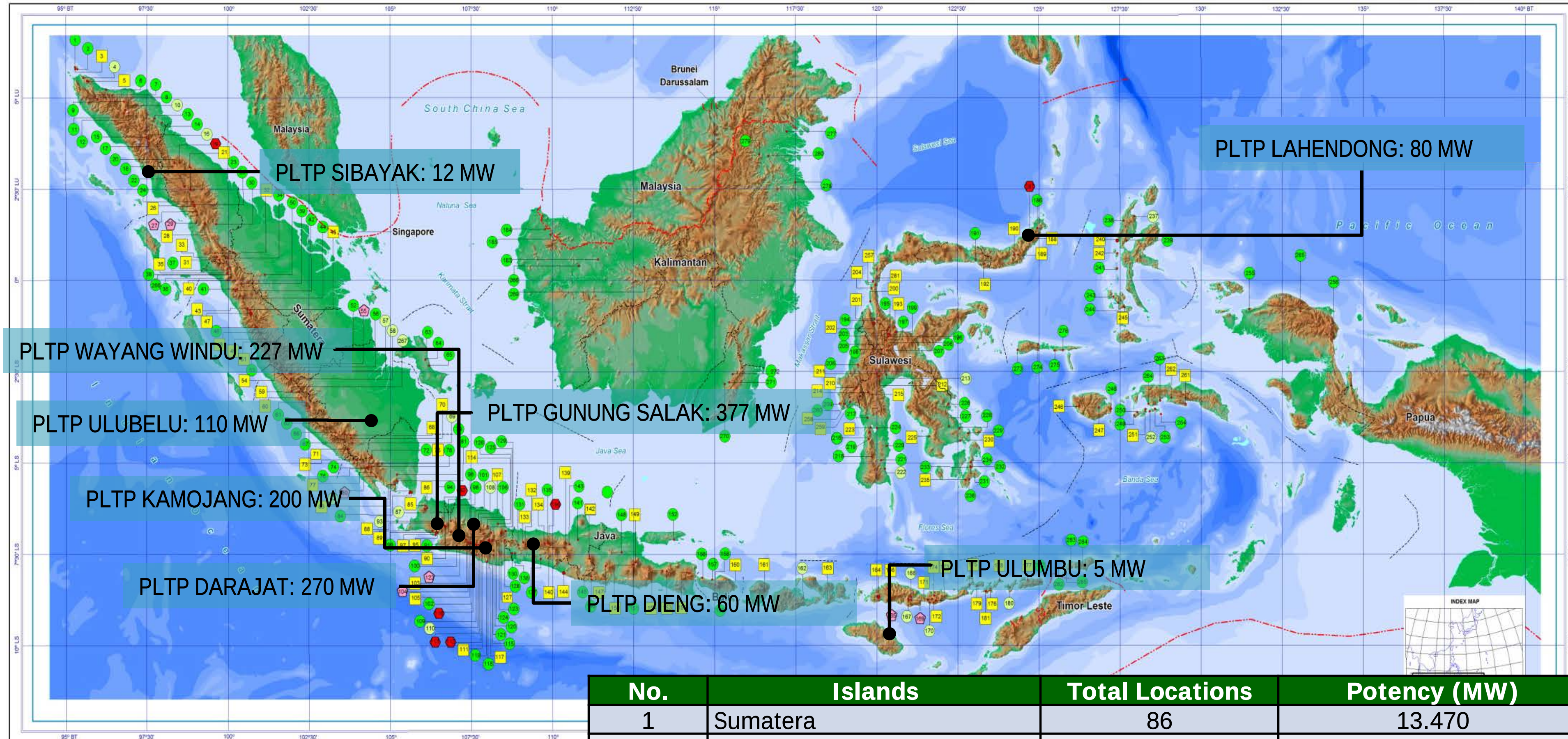
NEW AND RENEWABLE ENERGY RESOURCES

NO	TYPE	RESOURCE	INSTALLED CAPACITY (MW)	RATIO (%)
1	2	3	4	5 = 4/3
1	Hydro (MW)	75,000 MW	6,848.46 MW	9.13%
2	Geothermal (MW)	29,164 MW	1,341 MW	4.6 %
3	Biomass (MW)	49,810 MW	1,644.1 MW	3.3%
4	Solar Energy	4.80 kWh/m^2/day	22.45 MW	-
5	Wind Energy	3 – 6 m/s	1.87 MW	-
6	Ocean	49 GW [2)]	0.01 MW [3)]	0%
7	Uranium (MW)	3,000 MW	30 MW [1)]	0%

Source: Ministry of Energy and Mineral Resources , *reprocessed by NEC*

Note: 1) Research scale, non-energy
2) Source: National Energy Council
3) Research Scale: BPPT

GEOTHERMAL POTENTIAL



No.	Islands	Total Locations	Potency (MW)
1	Sumatera	86	13.470
2	Jawa & Bali	76	10.013
3	Nusa Tenggara	22	1.471
4	Kalimantan	12	145
5	Sulawesi	56	2.939
6	Maluku & Papua	33	1.126
TOTAL		**285**	**29.164**

Preliminary Survey

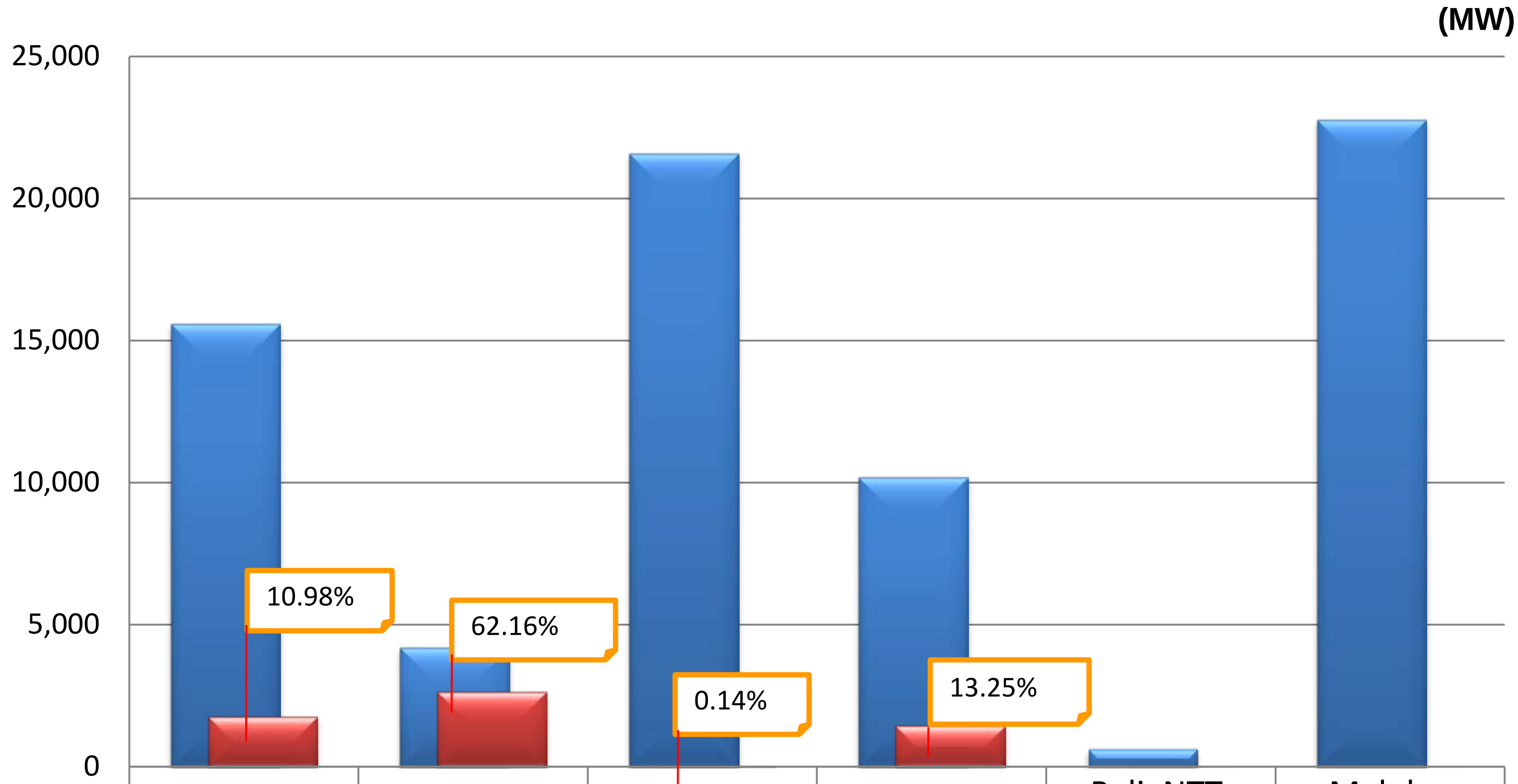
Ready to be developed

Detailed Survey

Installed

Source: Geological Agency, Ministry of Energy and Mineral Resources (2012)

HYDRO POTENTIAL AND INSTALLED CAPACITY

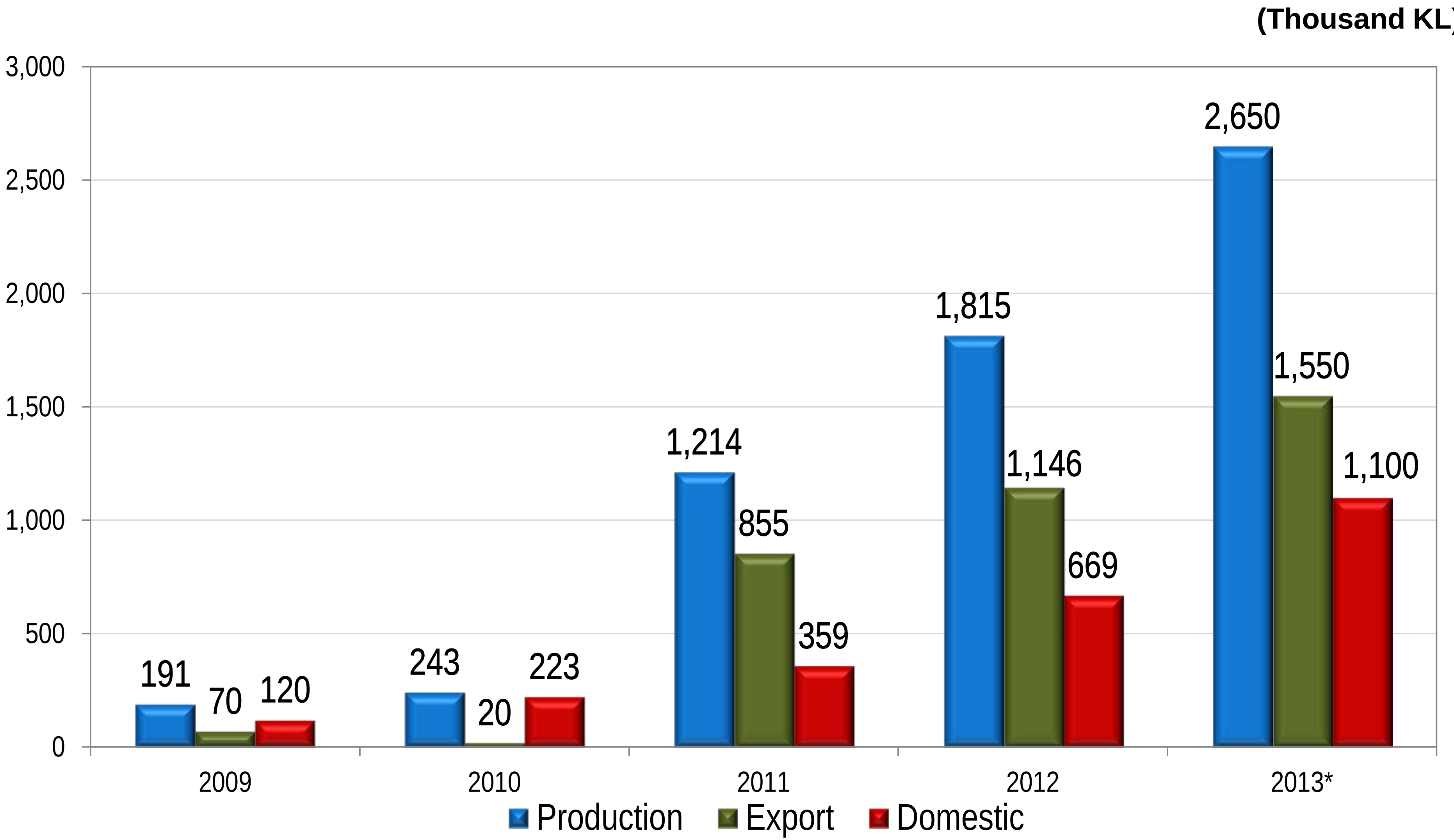


	Sumatera	Java	Kalimantan	Sulawesi	Bali, NTT Bali, Nusa Tenggara	Maluku, Papua
Potential	15,600	4,200	21,600	10,200	620	22,780
Installed Capacity	1,772.60	2,652.40	31.34	1,460.41	14.65	8.59

- Indonesia have 75 GW total hydropower potential and 5,940 MW installed capacity (7.9% of potential)
- The Largest hydropower potential is in Papua and Kalimantan, but its development is still constrained by the construction of hydropower infrastructure.
- Installed capacity data included minihydro and microhydro powerplant.

Source: Ministry of Energy and Mineral Resources (EBTKE,2012) , *reprocessed by NEC*

NATIONAL BIO-FUEL PRODUCTION



Source: Ministry of Energy and Mineral Resources (2012) , *reprocessed by NEC*

Note : *) tentative figure

Biodiesel Installed Capacity: 4.25 million KL, and bioethanol: 153 million KL

CO_2 Emissions

NATIONAL COMMITMENT TO REDUCE GHG

Presidential Regulation Number 61 Year 2011
to reduce greenhouse gasses (GHG) emission by 2020
(Indonesia Commitment in G-20 Pittsburgh and COP15)

Domestic effort

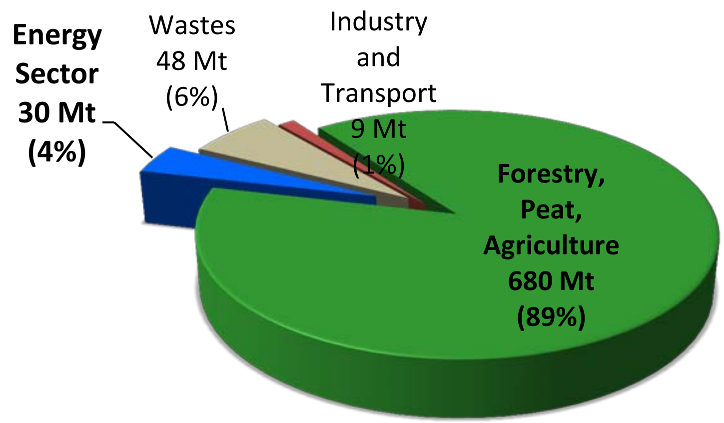
26%
(0.767 Giga Tonne CO_2e)

41%
(1.189 Giga Tonne CO_2e)

Domestic effort and international support

New Energy Development

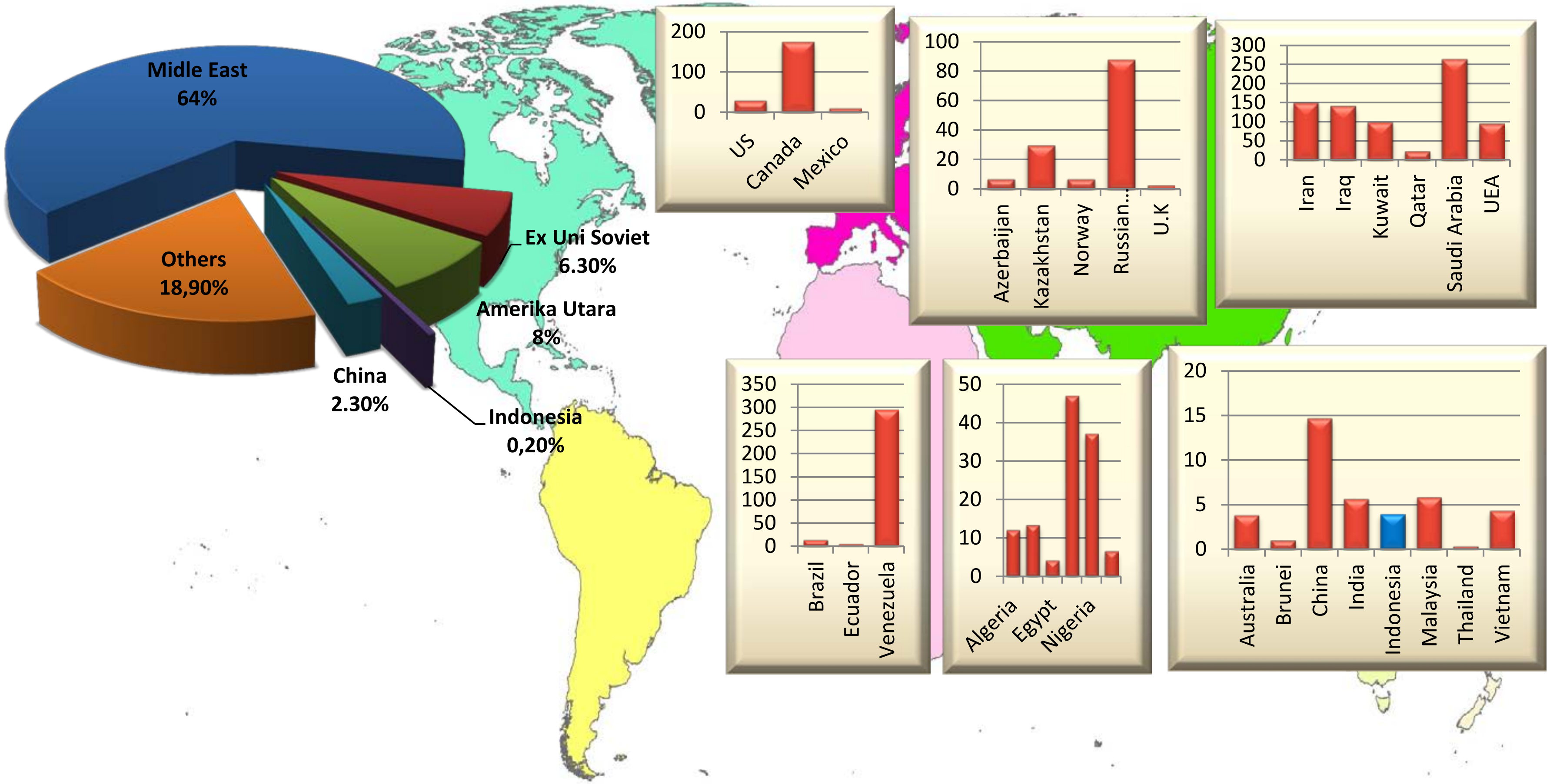
Energy Conservation



INDONESIA'S POSITION IN THE GLOBAL CONTEXT

WORLD OIL RESERVES

(Thousand MMbbl)

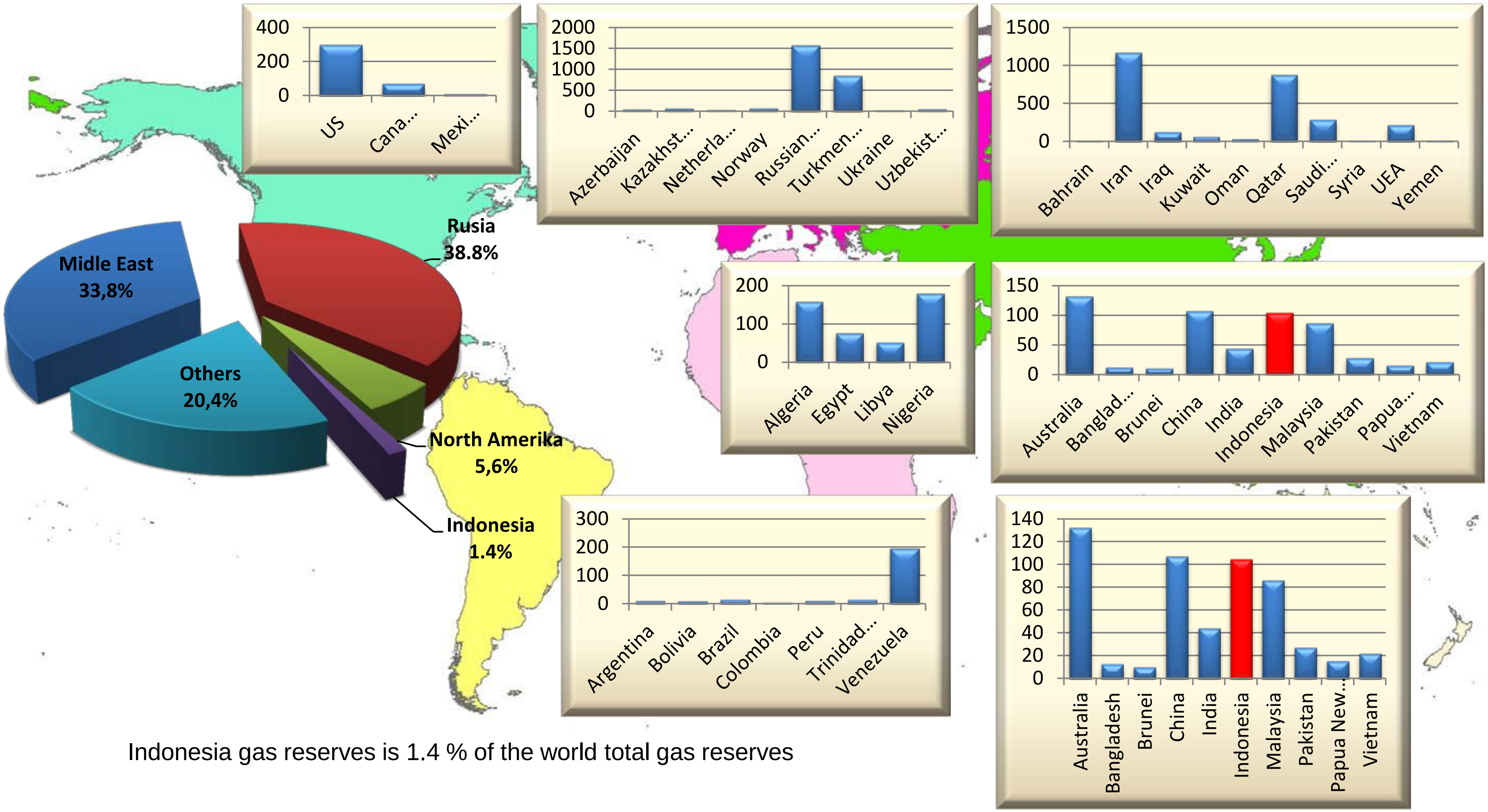


Indonesia Oil reserves is 0.2 % of the world total oil reserves

Source: BP Statistical Review, by 2012

WORLD NATURAL GAS RESERVES

(TCF)

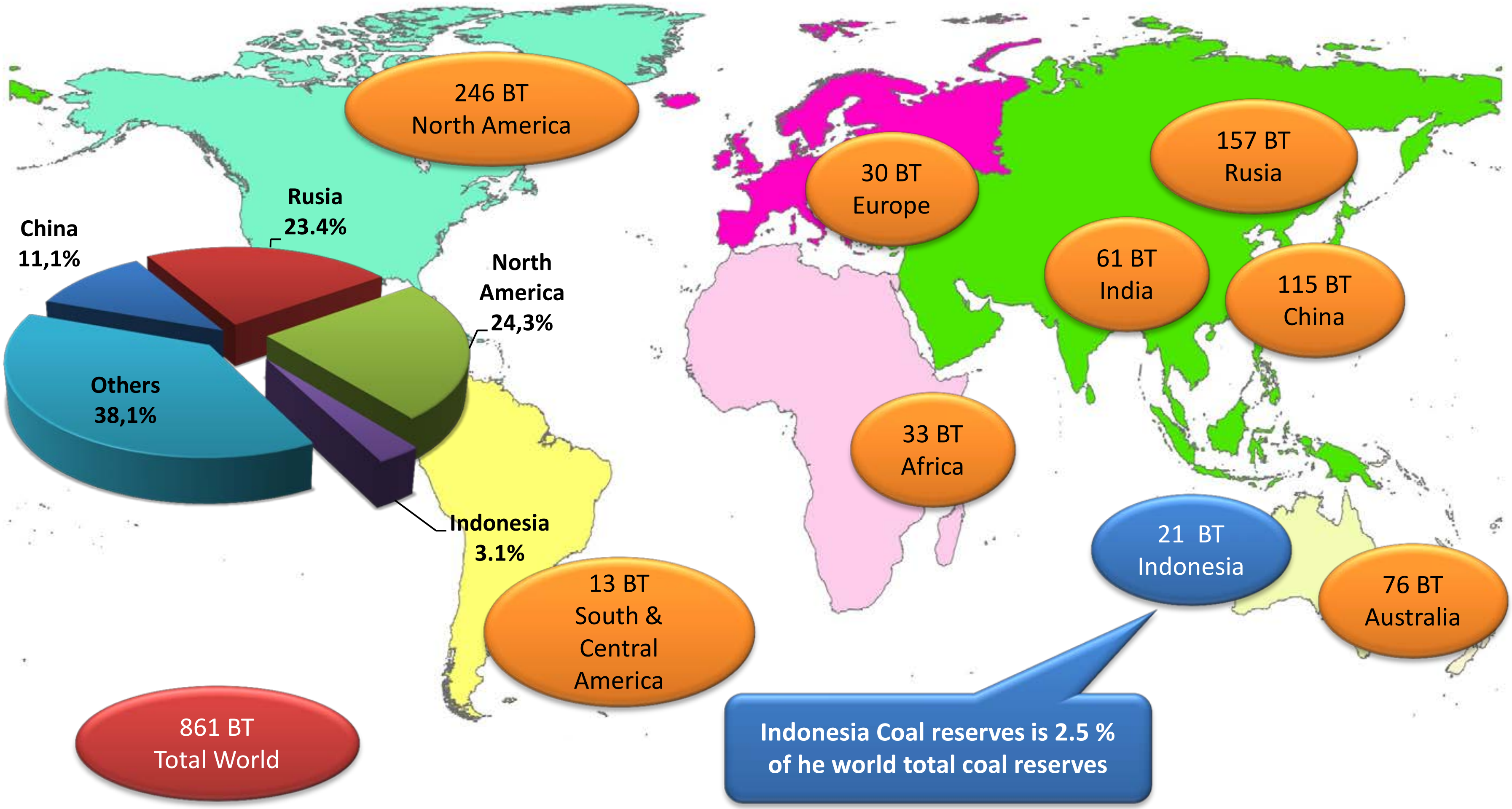


Indonesia gas reserves is 1.4 % of the world total gas reserves

Source: BP Statistical Review, by 2012

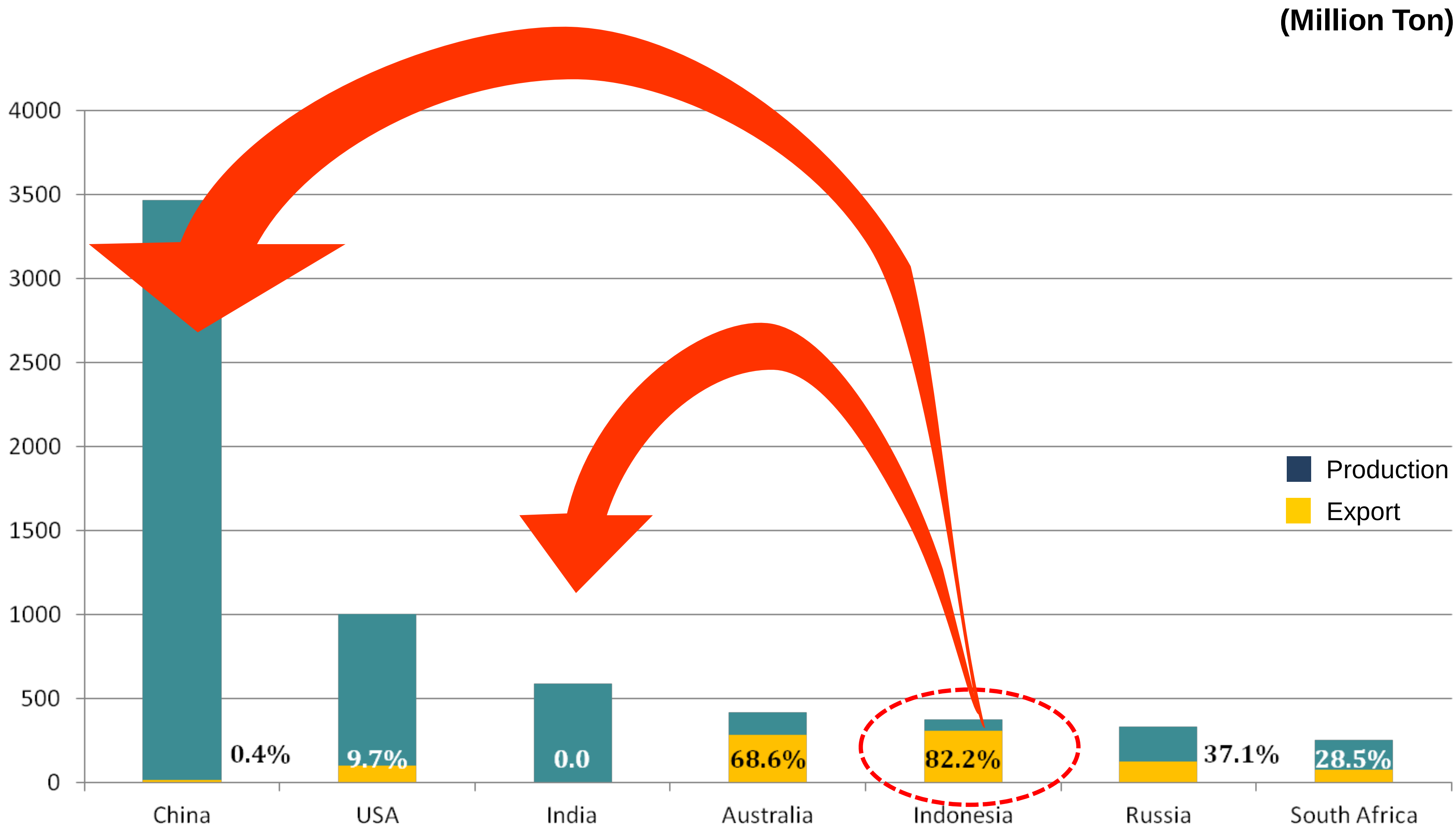
WORLD COAL RESERVES

Billion Ton



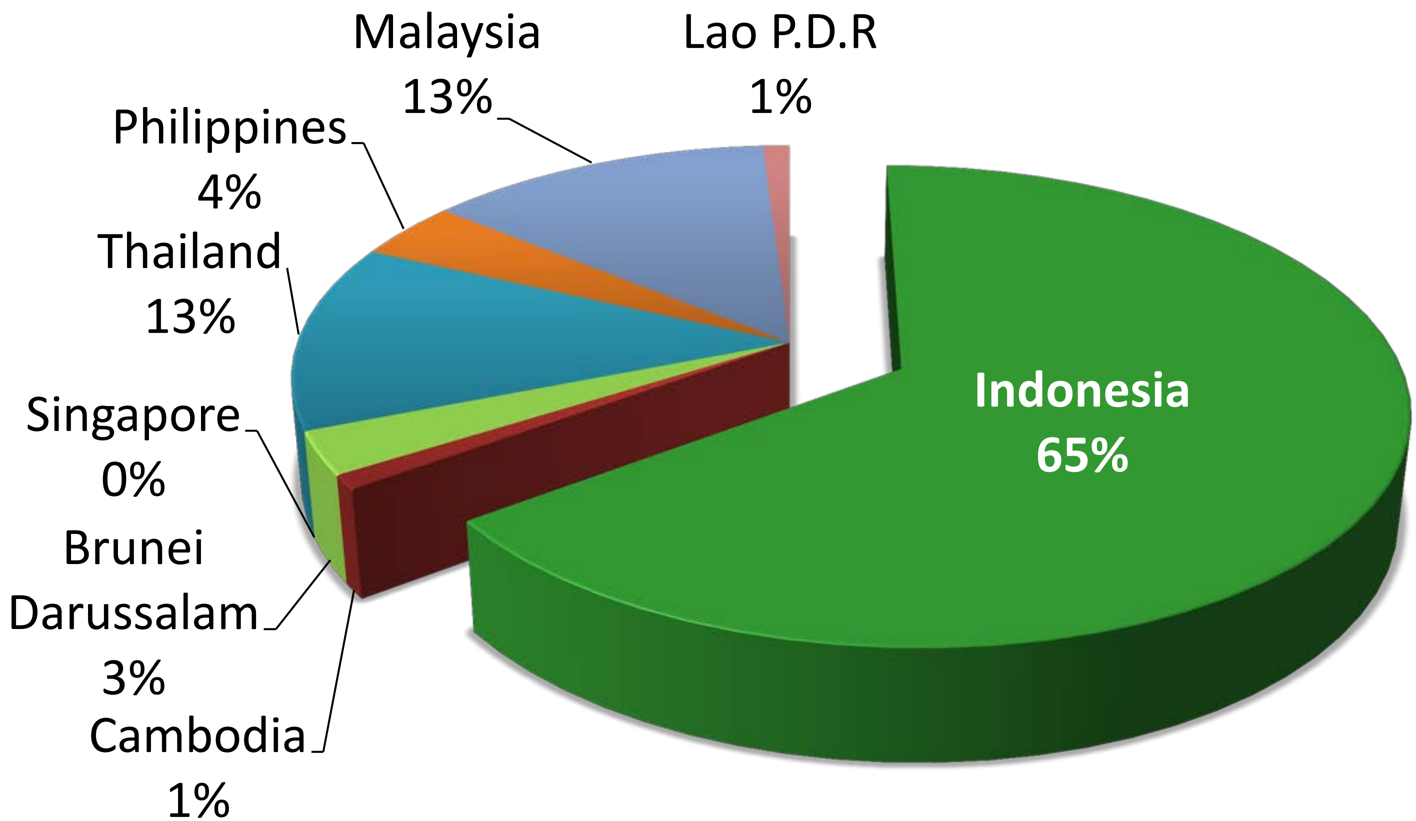
Source: BP Statistical Review, by 2012

COAL PRODUCTION-EXPORT COMPARISON



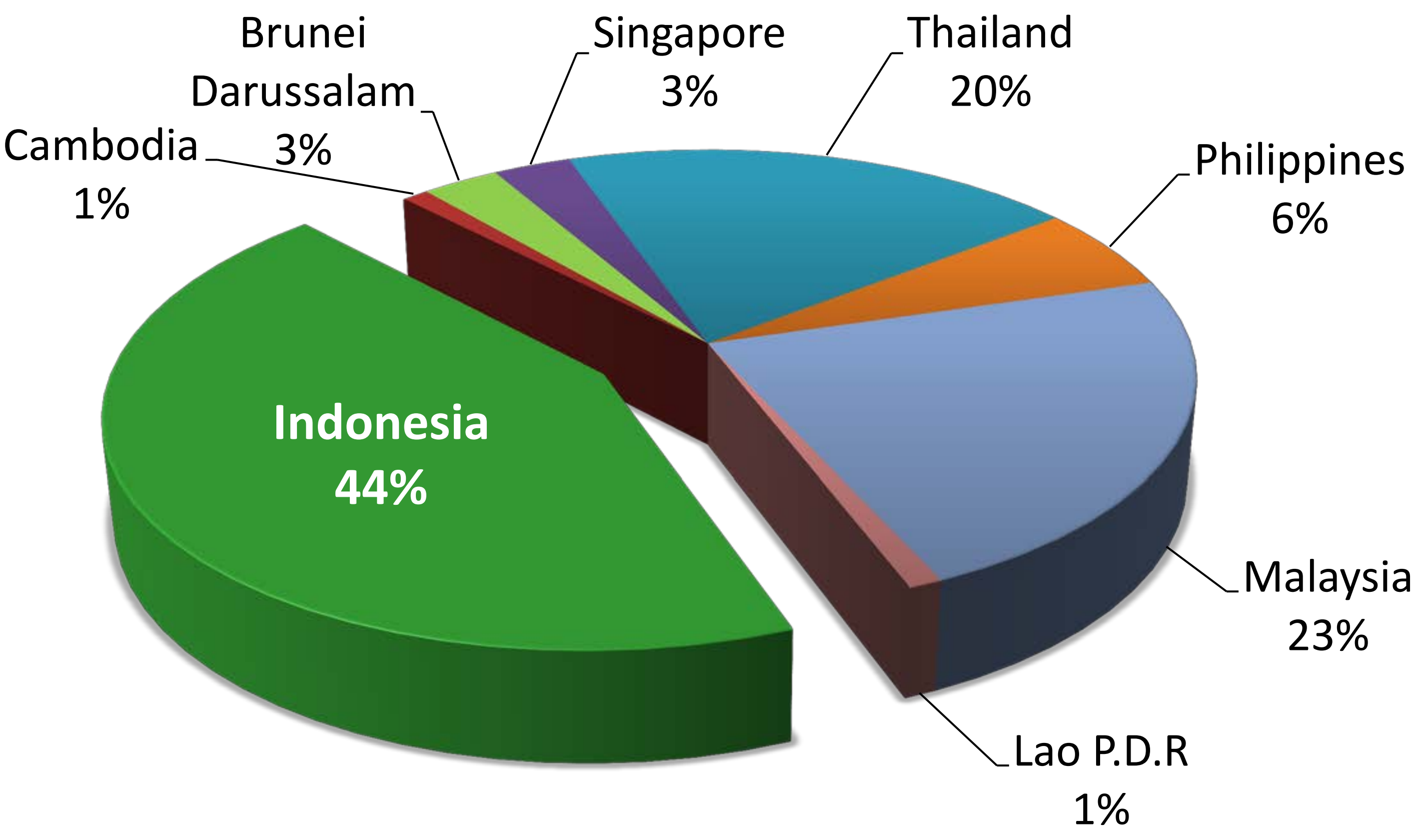
Note: In 2012, Indonesian coal exports to China was 92,15 million and to India was 64,371 million tons

SHARE OF ASEAN PRIMARY ENERGY PRODUCTION



Source: ASEAN Center for Energy, 2011
Note : Data of other ASEAN member states are not yet provided

SHARE OF ASEAN ENERGY DEMAND

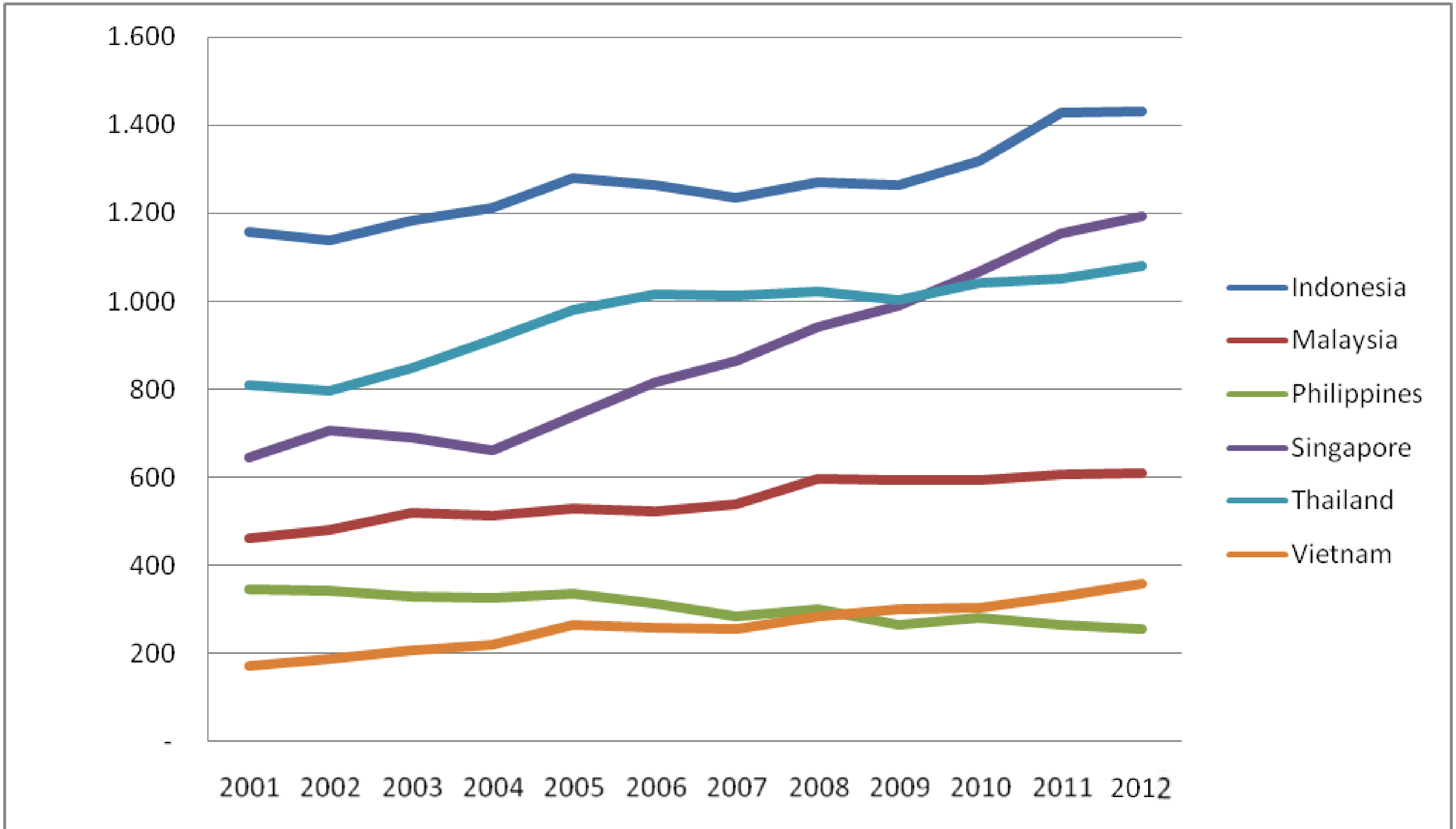


- Southeast Asia's energy demand increases by over 80% between 2013 and 2035 (IEA)
- Oil demand rises from 4.4 mb/d today to 6.8 mb/d in 2035, natural gas demand increases by 80% to 250 bcm

Source: ASEAN Center for Energy, 2011
Note : Data of other ASEAN member states are not yet provided

ASEAN OIL CONSUMPTION

(Thousand Barrel per day)

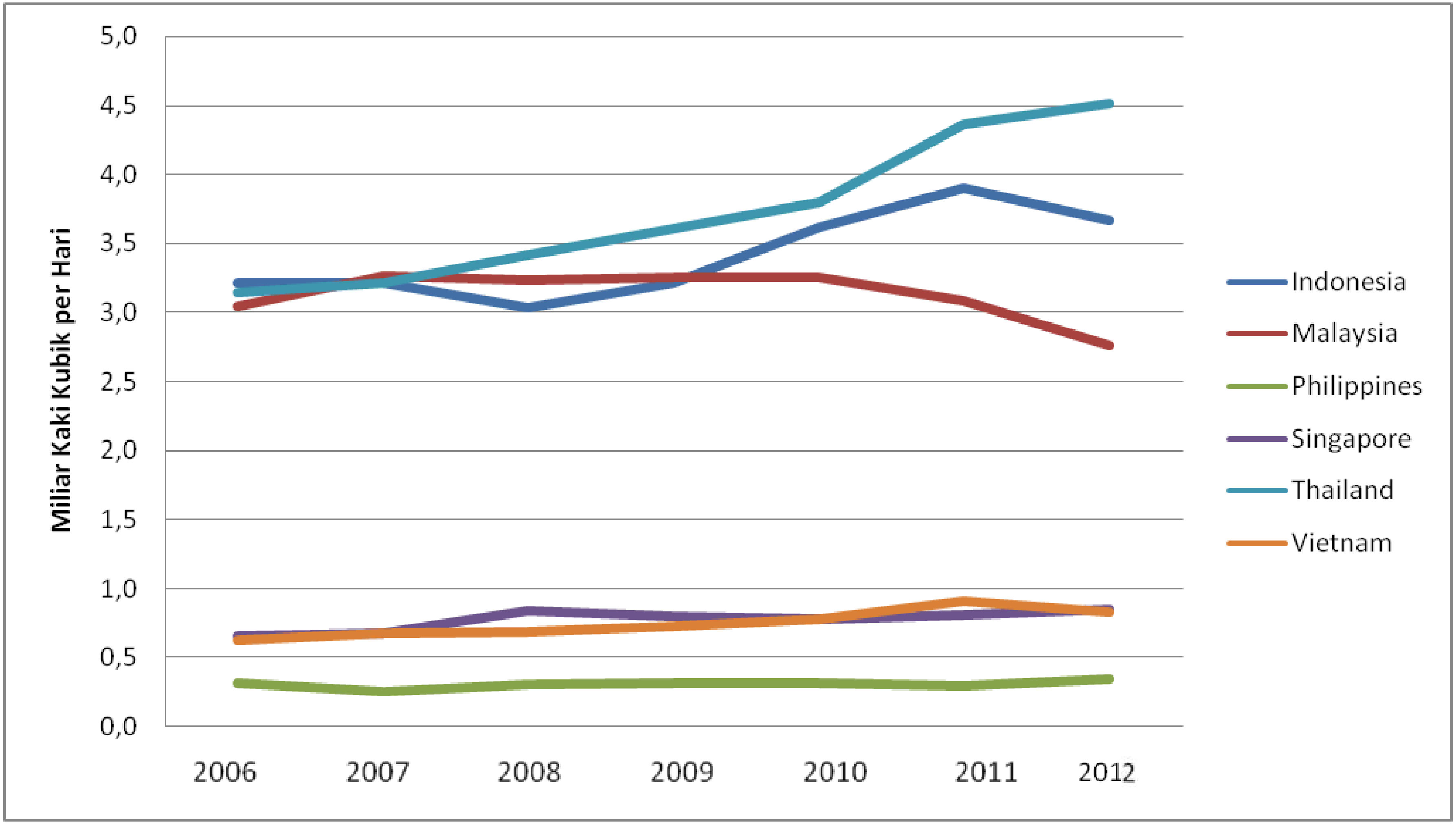


Source: BP statistical Review

Note : Data of other ASEAN member states are not yet provided

ASEAN GAS CONSUMPTION

(Thousand Barrel per day)



Source: BP statistical Review
Note : Data of other ASEAN Member States are not yet provided